MIL-R-21530D (WEPS)
26 February 1962
Superseding
MIL-R-21530C (WEPS)
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MILITARY SPECIFICATION

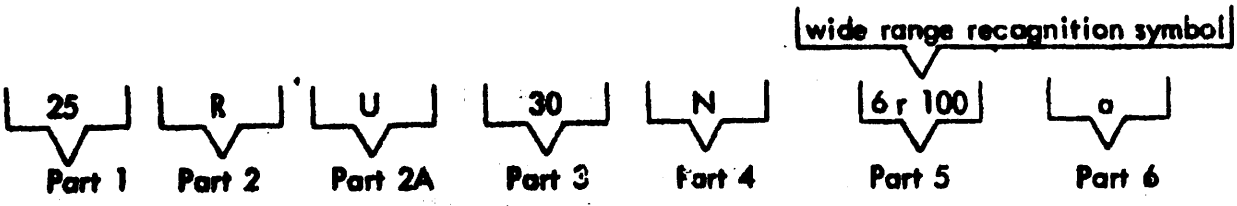
RESOLVERS, ELECTRICAL, AC
FOR USE IN
POLARIS FLEET BALLISTIC MISSILE WEAPON SYSTEM
GENERAL SPECIFICATION FOR

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. - This specification prescribes the requirements for the procurement of Resolvers, Electrical, AC (hereinafter termed resolvers), which will be used primarily at frequencies of 60 through 10,000 cycles per second. The resolvers specified herein are electromechanical devices which produce voltages on the secondaries which are sine and cosine functions of the voltages supplied to the primaries, depending on the angular position of the rotor with respect to the stator.

1.2 Type designations. - The type designation shall consist of a 6-part symbol, defined as follows:



1.2.1 Part 1. The digits to the left of the letter "R" designate the size of the resolvers in tenths of an inch to the nearest whole tenth above the maximum diameter.

FSC-5990

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1.2.2 Part 2. The letter "R" designates resolvers.

1.2.3 Part 2A. The letter "U" that may follow the letter "R" designates special mounting styles.

1.2.4 Part 3. The digit(s) following the letter "R" designates the input impedance in hundreds of ohms to the nearest whole hundred above the maximum impedance.

1.2.5 Part 4. The letter following part 3 designates the compensation as follows:

1.2.5.1 Resistor compensated: R.

1.2.5.2 Winding compensated: W.

1.2.5.3 Resistor and winding compensated: B.

1.2.5.4 Not compensated: N.

1.2.6 Part 5. The digit (s) following part 4 designates the excitation frequency as follows:

1.2.6.1 60 cps: 6.

1.2.6.2 400 cps: 4.

1.2.6.3 800 cps: 8.

1.2.6.4 1,000 cps: 10.

1.2.6.5 10,000 cps: 100.

1.2.6.6 Wide range: Resolvers designed to operate over a range of frequencies without any degradation of performance will have the range of excitation frequencies designated by using two code numbers defining the limits of the operating range, as selected from 1.2.6.1 through 1.2.6.5, with the letter "r" between the two code numbers.

1.2.7 Part 6. The small Latin letter that may conclude the symbol is a serial designation of the modification to the basic design as identified on the referenced drawings.

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Method of
MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-S-901	Shockproof Equipment, Class HI (High-Impact) Shipboard Application, Tests for
MIL-E-4970	Environmental Testing, Ground Support Equipment, General Specification for
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-I-16910	Interference Measurements; Radio, Methods and Limits, 14 kc to 1000 mc
MIL-P-16298	Preservation, Packaging, Packing and Marking of Electric Machines Having Rotating Parts and Associated Repair parts
MIL-P-21415	Protective Finish and Painting of Polaris Fleet Ballistic Missile
MIL-Q-21549	Quality Assurance Program Requirements for Fleet Ballistic Missile Weapon System Contractors

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-202	Test Methods for Electric and Electronic Component Parts
MS-33540	Safety Wiring, General Practices for
MS-33586	Metals, Definition of Dissimilar

PUBLICATIONS

Bureau of Naval Weapons

OD 23404	Considerations Involved in the Testing and Application of Electrical Resolvers
OP 400	Design, Manufacture, and Inspection of Naval Ordnance Equipment
OP 2230	Workmanship and Design Practices for Electronic Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. - The resolvers shall consist of electromechanical devices comprised of a stator and rotor. The stator and rotor shall each have two perpendicular (electrically) windings whose leads are brought out separately. Upon the occurence

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of an input voltage (to either the rotor or stator), the resolvers shall provide an output voltage (from either the stator or the rotor) which is a function of the resolution of the rectangular components of the input voltage and the mechanical (angular) position of the rotor, and shall transmit the angular data as specified herein for accuracy and other electrical characteristics thereof. The resolver shall approach, as specified herein, an ideal resolver as defined by the following equations:

(Symbols and terms are defined in section 6 herein.)

3.1.1 Rotor-energized resolver equations.

3.1.1.1 $E(S1 - S3) = KE\ (R1 - R3) \cos \theta - KE\ (R2 - R4) \sin \theta$.

3.1.1.2 $E(S2 - S4) = KE\ (R2 - R4) \cos \theta + KE\ (R1 - R3) \sin \theta$.

3.1.2 Stator-energized resolver equations.

3.1.2.1 $E(R1 - R3) = KE\ (S1 - S3) \cos \theta + KE\ (S2 - S4) \sin \theta$.

3.1.2.2 $E(R2 - R4) = KE\ (S2 - S4) \cos \theta - KE\ (S1 - S3) \sin \theta$.

3.2 General requirements.

3.2.1 Preproduction samples. - Unless otherwise specified in the contract or purchase order, preproduction samples of the resolver shall be manufactured using the methods and procedures proposed for the production lot. These samples shall be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification.

3.2.2 Materials and purchased parts. - Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of the resolvers shall conform to the detailed requirements stated herein, indicated on drawings and specified in applicable documents as listed herein. With the exception of specific requirements herein, all materials and purchased parts entering into the final assembly of the resolvers shall be selected in accordance with OP 400.

3.2.2.1 Flammable materials. - Materials which may be ignited or cause an explosion shall not be used for construction applications where sparking or arcing could cause igntion. Materials which may inadvertently support combustion shall

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not be used. Flammable impregnating, potting or embedding compounds, when enclosed in fire-resistant containers, shall be acceptable.

3.2.2.2 Toxic materials. - Materials which produce harmful toxic effects during service usage shall not be used. Where a suitable nontoxic materials is not available, preventive measures shall be provided, subject to approval of the procuring activity. Materials which, when ignited, produce toxic substances or vapors shall be avoided whenever practicable.

3.2.2.3 Unstable materials. - Materials which are or become unstable to the extent that they emit or deposit substances capable of causing corrosion or producing electrical leakage paths within an assembly under possible operating and storage conditions shall not be used during manufacture or recommended for use during maintenance.

3.2.2.4 Dissimilar metals. - The use of dissimilar metals, as defined in MS 33586, shall be avoided whenever practicable. In the event dissimilar metals are used, all the requirements of this specification shall be met. Magnesium shall not be used except with the prior approval of the procuring agency or when so called for on the applicable drawings.

3.2.3 Finishes and coatings. - The finishes and coatings of the resolvers shall be in strict accordance with the applicable drawing. All exterior metal surfaces shall be treated in accordance with MIL-P-21415. Plastics and other materials used in the construction of the resolvers shall be finished in accordance with the applicable material specification.

3.2.4 Construction. - The design and construction of the resolvers shall be in accordance with OP 400. Designs shall be submitted to the procuring activity prior to fabrication. The design and construction of the resolver shall not be changed after acceptance and approval of the preproduction sample, or of the first production lot, without prior approval of the procuring activity.

3.2.4.1 Mechanical construction. - The construction of the resolvers shall be such as to insure that the requirements as stated herein are met in all particulars. Where required, and as determined by the Government inspector, radiographic, visual, chemical or optical inspection of the fabrication, construction, joining and workmanship shall be accomplished to prove that no flaws, cracks, separation of laminations, or other impediments to full performance exist.

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3.2.4.2 Electrical construction. - Wiring, soldering and other mechanical details of electrical construction shall conform in general to OP 2230.

3.2.5 Moisture, fungus and corrosion treatments. - The resolvers shall be rendered moisture, fungus and corrosion resistant in accordance with the detail requirements specified for each material used in the construction of the equipment and as follows:

3.2.5.1 Fungus treatment. - Components and materials with insufficient moisture or fungus resistance shall be given appropriate fungus treatment. Such agents shall be included in potting compounds if the compound is susceptible to fungus. Materials and parts shall be treated before assembly if at all practicable. If treatment is not possible before assembly, the requirements of MIL-T-152 shall apply. Types and concentrations of fungistatic or fungicidal materials shall be approved by the procuring activity.

3.2.5.2 Protection from corrosion. - Metals not inherently resistant to corrosion, but which are appropriate for use, shall be given protection by painting, plating with a corrosion resistant or otherwise suitable metal, a suitable surface treatment, or a combination of such processes in accordance with BuWeps approved specifications. Metals used in electrical connections shall have characteristics such that the electrical conductivity of the connection shall not change to the extent that the electrical performance is affected during periods of storage, operation or test over the specified environmental range.

3.2.6 Form factor. - The final over-all dimensions and weight of the resolvers shall be in accordance with the applicable drawing.

3.2.7 Assembly and components. - All components of the resolvers which can be assembled in an ambiguous manner shall be furnished with such assembly guides, keys or markings as to insure assembly in the correct relative position only.

3.2.8 Gages. - The contractor shall provide himself with whatever gages are necessary in accordance with OP 400.

3.2.9 Markings. - The resolvers shall have markings as follows:

3.2.9.1 Nameplate. - The information as shown on the applicable drawing for each resolver shall be marked in accordance with MIL-STD-130.

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3.2.9.2 Terminal identification. - Those resolvers having screw or lug type terminals shall have terminal identification silkscreened, molded, etched, or stamped in contrasting waterproof ink or paint at or near the respective terminal as follows:

(a) Stator winding: S1, S2, S3, S4.

(b) Rotor winding: R1, R2, R3, R4.

(c) Compensating winding: C1, C2, C3, C4.

(d) Compensating resistor: 1, 2, 3, 4.

Terminal designation shall be such that stator and compensator terminals with like numbers have like polarity.

3.2.9.3 Other markings. - All other markings that the contractor deems necessary shall be submitted to the procuring activity for approval prior to application to the resolvers.

3.2.10 Color coding

3.2.10.1 Lead wire. - For units using lead wires instead of terminals, the wire color code designation shall be in accordance with table I.

Table I

Rotor		Stator		Compensator	
Terminal	Color	Terminal	Color	Terminal	Color
R1	Red	S1	Red	C1	Red and white tracer
R2	Yellow	S2	Yellow	C2	Yellow and white tracer
R3	Black	S3	Black	C3	Black and white tracer
R4	Blue	S4	Blue	C4	Blue and white tracer

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3.2.10.2 Sleeving. - A red sleeve shall be used for rotor leads and a black sleeve shall be used for stator leads, except in cases where the leads are easily discernable.

3.2.11 Manufacturing and interchangeability. - Resolvers shall be manufactured in accordance with the requirements of this specification and the applicable documents for each type of resolver. All resolvers having identidical type designations shall be mechanically and electrically interchangeable within all performance requirements and product characteristics specified herein.

3.2.12 Requirements for individual types of resolvers. - All requirements specified herein apply to all types of resolvers covered by this specification. Detail requirements of exceptions applicable to individual types are specified on the applicable drawing or in the detail specification for each resolver type.

3.2.13 Matched sets of components. - Resolvers covered by this specification which are supplied in their component parts (i. e., disassembled) shall, within the requirements of interchangeability, be supplied as matched sets of the components required to assemble a complete resolver. A matched set of components shall be that set of components that were assembled together as a resolver (in accordance with the direction of the applicable drawing) in order to be tested to meet the requirements of this specification.

3.2.13.1 Set identification. - Those resolvers supplied as disassembled parts shall have each part stamped with a number suitably compounded to permit ready identification of the component part as belonging to a matched set of such components as shall assemble into one resolver in accordance with 3.2.13.

3.3 Reliability. - The resolver shall be able to operate at a maximum failure rate of 5 units every 1,000,000 hours, or 2,000 consecutive hours of operation without maintenance and a probability of failure of 1.0 percent.

3.4 Performance requirements and product characteristics. - The resolvers shall meet the following performance requirements and product characteristics.

3.4.1 Input voltage and frequency. - The resolvers shall operate as specified herein over the input voltage and frequency range supplied to the primary as specified in the applicable detail specification.

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3.4.2 Input voltage harmonics. - The resolvers shall operate as specified herein when energized from a power source having not more than 2.0 percent total harmonic distortion and no more than 1.0 percent at any single harmonic component over the entire input voltage range of 3.4.1.

3.4.3 Test voltage. - The test voltage for each resolver shall be as specified in the detail specification.

3.4.4 Impedance. - The resolver, when tested in circuit A1 of appendix, shall exhibit impedances as follows:

3.4.4.1 Zro. - The impedance of each rotor winding, when measured separately at the test voltage and test frequency with the remaining windings open-circuited, shall be as specified in the detail specification.

3.4.4.2 Zso. - The impedance of each stator winding, when measured separately at the test voltage and test frequency with the remaining windings open-circuited, shall be as specified in the detail specification.

3.4.4.3 Zrs. - The impedance of each rotor winding, when measured separately at the rated current with the stator windings shorted, shall be as specified in the detail specification.

3.4.4.4 Zss. - The impedance of each stator winding, when measured separately at the rated current with the rotor windings short-circuited, shall be as specified in the detail specification.

3.4.4.5 Zco. - The impedance of each compensator winding, when measured separately at the test voltage and test frequency with the remaining windings open-circuited, shall be as specified in the detail specification.

3.4.5 Resolver zero marking. - The resolvers shall have coarse resolver zero markings to correspond to electrical zero, as defined in 6.4.1.16, with the accuracy specified herein.

3.4.5.1 Rotor excited units. - The marking on the stator shall be within ± 10 degrees of the marking on the rotor when the resolver is tested in the circuit shown in figure 1 and minimum voltage is obtained on the voltmeter.

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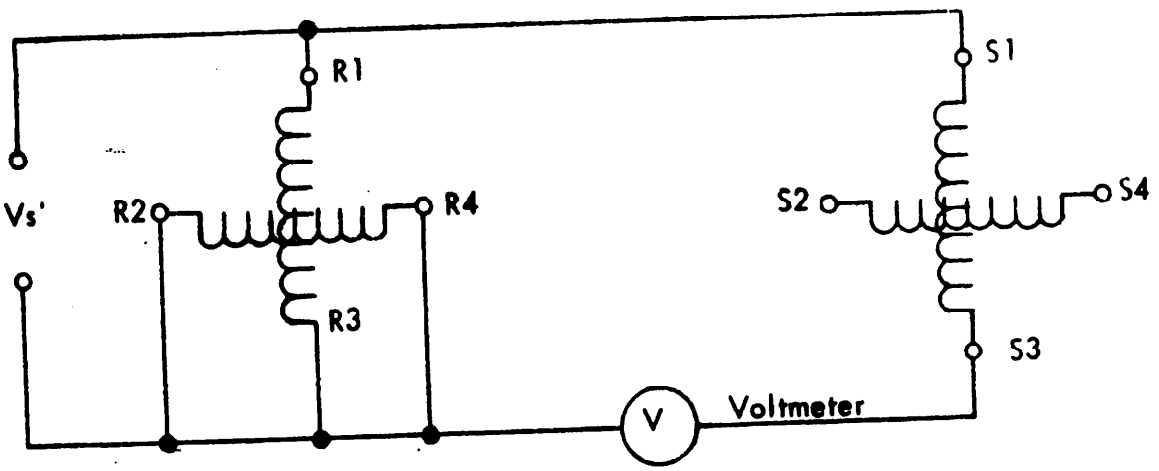


Figure 1

3.4.5.2 Stator excited units. - The marking on the stator shall be within ± 10 degrees of the marking on the rotor when the resolver is tested in the circuit shown in figure 2 and minimum voltage is obtained on the voltmeter.

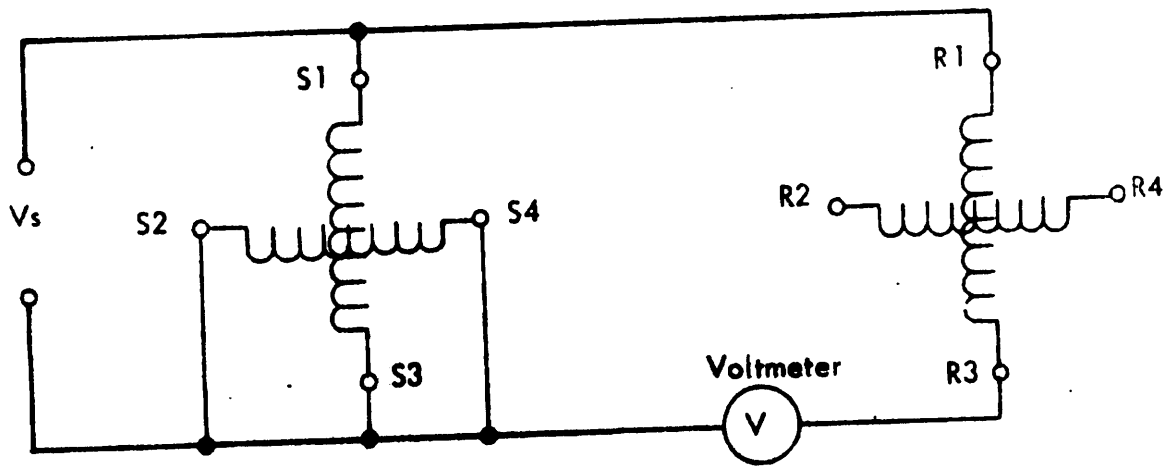


Figure 2

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3.4.6 Phase rotation. - The resolvers shall have outputs in accordance with the equations of 3.1.1 and 3.1.2 for positive rotation when tested as follows:

3.4.6.1 Rotor excited units. - With the unit connected as shown in figure 3 and 3A respectively, and when the rotor is rotated in a counterclockwise (ccw) direction from zero, the voltmeter reading shall decrease during the first 90 degrees of rotation.

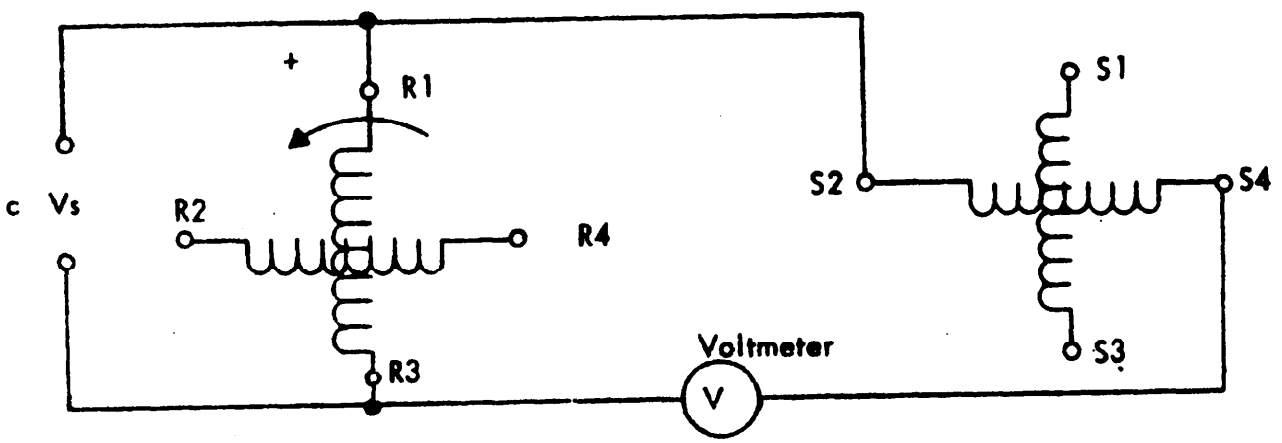


Figure 3

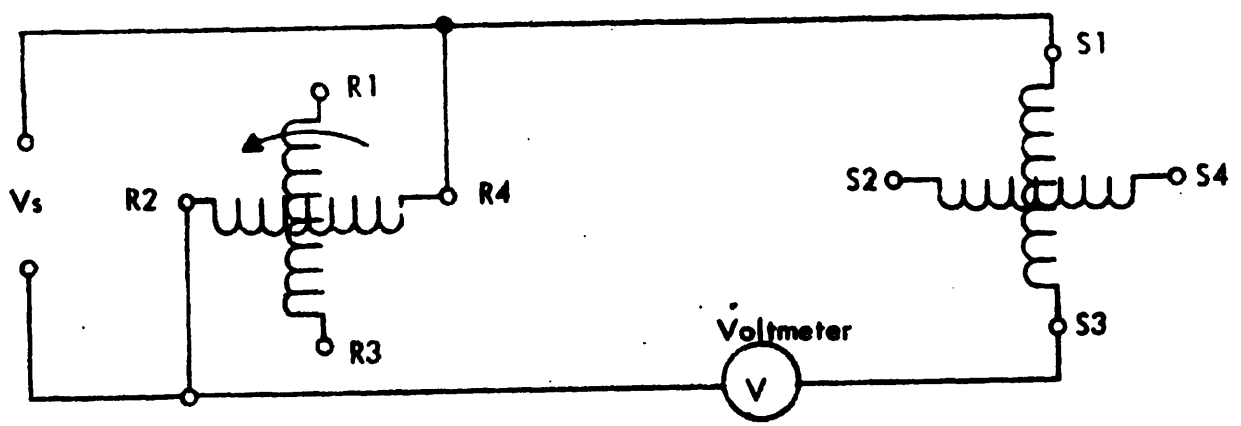


Figure 3A

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3.4.6.2 Stator excited units. - With the unit connected as shown in figure 4 and 4A respectively, and when the rotor is rotated in a ccw direction from zero, the voltmeter reading shall decrease during the first 90 degrees of rotation.

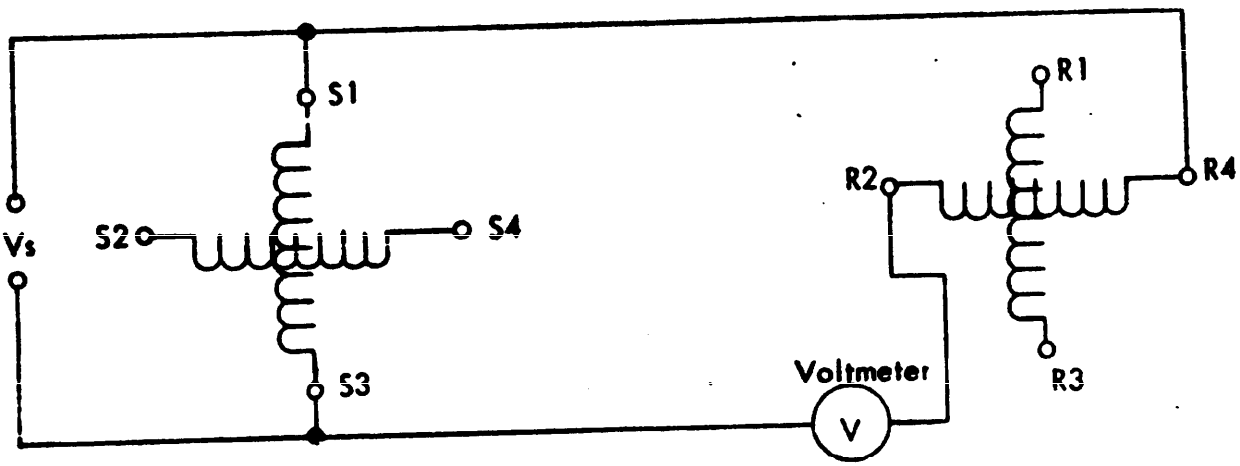


Figure 4

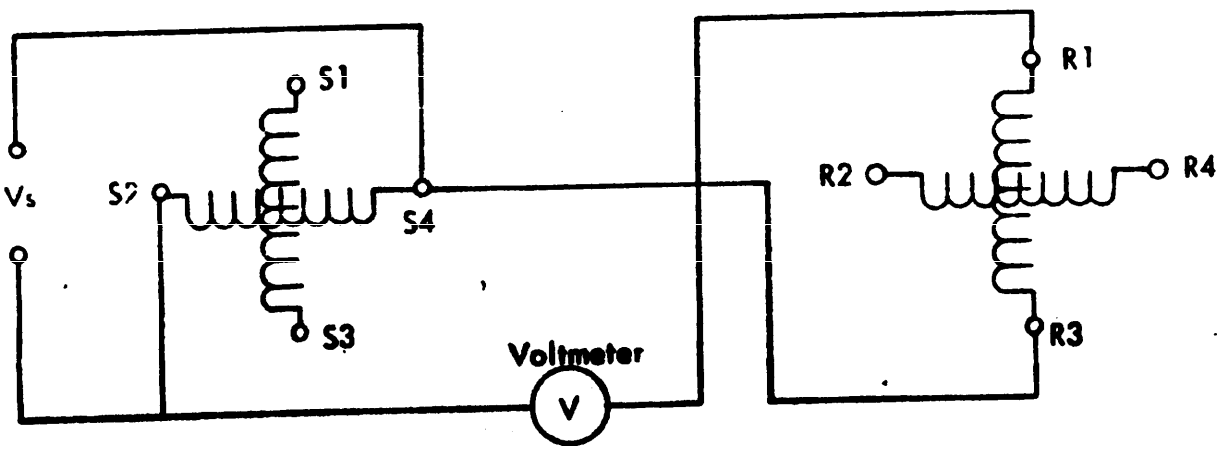


Figure 4A

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3.4.7 Harmonic distortion. - Unless otherwise specified in the applicable detail specification, the harmonic distortion of the resolvers shall be less than 0.5 percent when tested in accordance with the following:

3.4.7.1 Rotor excited units. - When tested in the circuit of figure 5 and measured at the 90 degree position, the harmonic distortion shall not exceed the value specified in the detail specification.

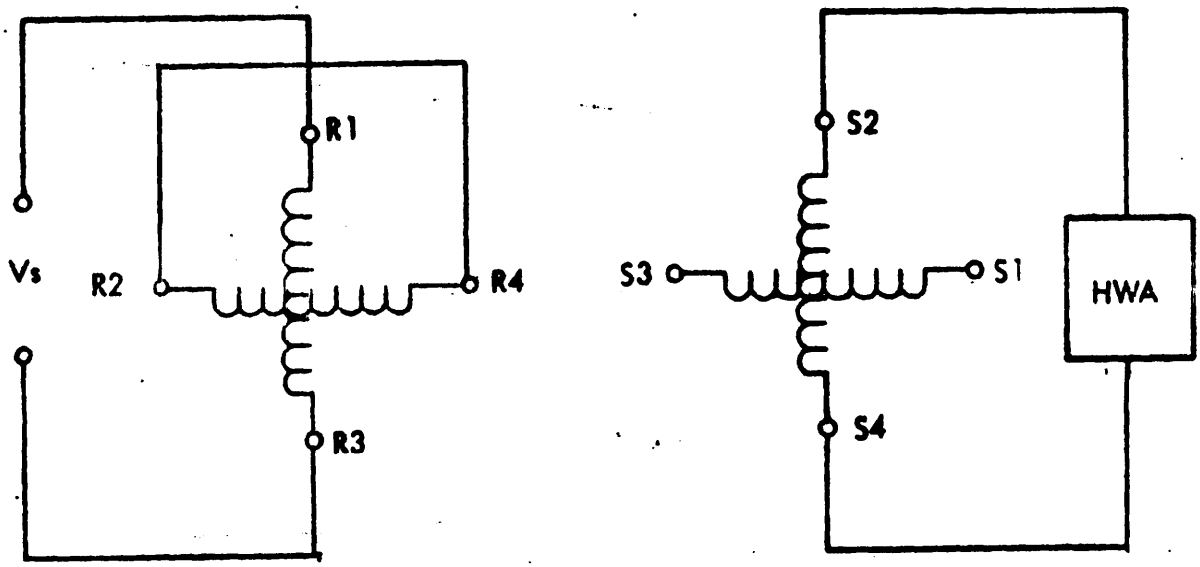


Figure 5

3.4.7.2 Stator excited units. - When tested in the circuit of figure 6, and measured at the 90 degree position, the harmonic distortion shall not exceed the value specified in the detail specification.

3.4.8 High potential (leakage resistance). - The resolvers shall show no more than 1.0 milliampere leakage current between windings and frame and between isolated windings when subjected to the following test voltages, applied from zero volts at the rate of 100 volts per second, and maintained thereat for 1 minute in an atmosphere the relative humidity of which is greater than 50 percent.

3.4.8.1 Primary excitation voltage greater than 50v rms. - Resolvers having an input voltage greater than 50v rms shall conform to the requirements of 3.4.8 when subjected to 60-cps test voltages of 900 v rms between windings and frame, and 450 vrms between isolated windings.

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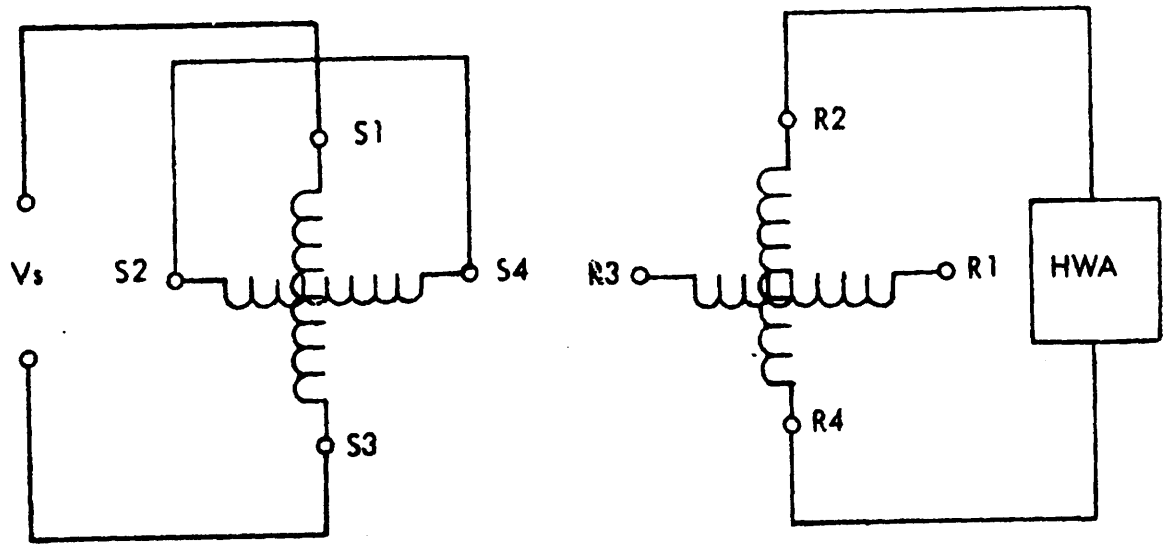


Figure 6

3.4.8.2 Primary excitation voltage less than 50v rms. - Resolvers having an input voltage less than 50 v rms shall conform to the requirements of 3.4.8 when subjected to 60-cps test voltages of 500 v rms between windings and frame, and 250 v rms between isolated windings.

3.4.8.3 Compensator windings. - Compensator windings shall conform to the requirements of 3.4.8 when subjected to a voltage of 300 v rms, 60 cps, between isolated windings and between windings and frame.

3.4.8.4 Testing derating. - The resolvers shall be tested at the full voltage specified herein for high potential tests for initial preproduction and periodic testing and at 80 percent of the specified voltages for acceptance testing and subsequent preproduction and periodic tests.

3.4.9 Insulation resistance. - The resolvers shall have an insulation resistance as follows when tested with a megohm bridge at a d-c test voltage of 500 v immediately following the high potential test of 3.4.8.

3.4.9.1 Size 15 and over. - Resolvers size 15 and over, when subjected to the test of 3.4.9, shall have an insulation resistance of at least 100 megohms between isolated windings and between windings and frame.

3.4.9.2 Under size 15. - Resolvers under size 15, when subjected to the test of 3.4.9, shall have an insulation resistance of at least 10 megohms between isolated windings and between windings and frame.

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3.4.10 Internal capacities. - If applicable, the principle equivalent internal capacities of each resolver shall be as specified in the applicable detail specification when tested in accordance with 4.8.22.

3.4.11 Total null voltage. - The total null voltage at any angle, as measured by an average-responding a-c voltmeter calibrated to read rms sinusoidal voltages, shall not exceed the specified limiting value of the detail specification for the temperature specified. The null voltage shall be read at each increment of angle with the null meter set to read total volts during angular accuracy test of 4.7.3.3.

3.4.12 Zero shift of null spacing at 45 degrees, angular error versus voltage. - After the resolver is set to 45 degrees from zero, the angular shift shall not exceed the specified limiting value of the detail specification when the input voltage is varied from the specified nominal test voltage to plus or minus 5.0 percent (± 0.5 percent) from the specified nominal test voltage.

3.4.13 Zero shift of null spacing at 45 degrees, angular error versus frequency. - After the resolver is set to 45 degrees from zero, the angular shift shall not exceed the specified limiting value of the detail specification when the frequency of the test voltage is varied plus or minus 5.0 percent from the nominal value.

3.4.13.1 Phase shift variation due to voltage. - The phase shift variations due to voltage when measured at the test voltages specified in 3.4.12 and in accordance with the methods outlined in 3.4.15 shall not exceed the limiting value of the detail specification.

3.4.13.2 Phase shift variations due to rotation. - The phase shift variations due to rotation, when measured in accordance with 4.8.21, shall not exceed the value specified in the detail specification.

3.4.14 Transformation ratio. - When the resolvers are tested in accordance with 4.7.4.3.1 herein the transformation ratio shall not exceed the specified limits of the detail specification.

3.4.15 Phase shift. - The phase shift of the resolvers shall not exceed the tolerance in the detail specification when tested in accordance with 4.7.4.3.2.

3.4.16 Angular error. - The angular error of the resolvers shall not exceed the specified value in the detail specification when tested in accordance with 4.7.4.3.3.

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3.4.17 Impedance balance. - The impedance unbalance between input or output windings of the resolver (which affects the accuracy of cascaded resolvers used for data transmission) shall not exceed the specified value in the detail specification when tested in accordance with 4.7.4.3.4.

3.4.18 Compensating windings, null voltage. - The null voltage of the compensating windings of the resolvers shall have a fundamental frequency component no greater than 0.10 millivolt rms per volt rms input when tested in accordance with 4.7.4.4.1.

3.4.19 Compensating windings, winding phase. - The correct winding phase of the compensating windings of the resolvers shall be indicated by a voltmeter reading less than the test voltage when the resolvers are connected in the circuit of and tested in accordance with 4.7.4.4.2.

3.4.20 Compensating windings, equality of. - The null voltage of the compensating windings of the resolvers shall have a fundmanetal frequency component no greater than 1.0 millivolt rms per volt rms input when tested in accordance with 4.7.4.4.3.

3.4.21 Compensating resistors. - Compensating resistors shall conform to the requirements of the applicable detail specification.

3.4.22 Radio interference. - When tested in accordance with the conditions of 4.8.7, the resolvers shall not produce radio noise at a distance of 3 feet from the unit in excess of 2.5 microvolts in the frequency range of 65 to 150 megacycles or shall not produce radio interference in the frequency spectrum from 15 kc to 1000 mc in excess of the limits in microvolts specified in MIL-I-16910.

3.4.23 Friction torque. - When tested in accordance with the requirements of 4.7.4.3.6 or equivalent method, the amount of torque required to overcome friction shall not exceed the values at the temperatures specified in the detail specification.

3.4.24 Brush-slip ring continuity. - When tested in accordance with 4.7.4.3.5, each rotor winding shall show continuity, and there shall be no change in resistance in excess of 0.5 percent or 1.0 ohm (whichever is greater) between the high and the low measured values unless otherwise specified in the detail specification.

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3.4.25 Temperature rise. - The temperature rise of the resolvers shall not exceed the specified values in the applicable detail specification when tested in accordance with 4.8.14.

3.5 Environmental requirements. - The resolvers shall meet the applicable requirements of this specification in accordance with table 2 (section 4) during or after the test specified in section 4 for the environmental requirements of this section.

3.5.1 Vibration, operating condition. - The resolvers shall withstand the vibration requirements specified in MIL-STD-167, type 1; and of method 204, condition B of MIL-STD-202; and as may be further specified in the applicable detail specification.

3.5.2 Vibration, shipping condition. - The resolvers, packed and packaged in accordance with any level outlined in section 5 herein, shall be capable of withstanding the vibration requirements of MIL-E-4970, category B, procedure III.

3.5.3 Shock, operating conditions. - The resolvers shall withstand the shock requirements of MIL-S-901 unless otherwise specified in the detail specification.

3.5.4 Shock, shipping condition. - The resolvers, packed and packaged in accordance with any level of section 5 herein, shall be capable of withstanding the shock requirements of MIL-E-4970, procedure III.

3.5.5 Tropical fungus environment. - The resolvers shall be capable of withstanding exposure to the fungus growth encountered in tropical climates as determined by testing in accordance with MIL-E-5272, procedure I, and thereafter shall show no damage.

3.5.6 Salt water (corrosion). - The resolvers shall be capable of withstanding exposure to the salt atmosphere encountered in or near sea environments as determined by testing in accordance with method 101 of MIL-STD-202 for 96 hours; and thereafter shall show no damage.

3.5.7 Storage life. - The resolvers shall be operable under any of the environmental conditions specified after having been stored for a period of at least 3 years when preserved, packed, and packaged to level A pack requirements as specified in section 5 herein under temperature and humidity conditions as follows:

3.5.7.1 Temperature. - The storage temperature shall be -54 to +71 degrees centigrade (C).

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3.5.7.2 Humidity. - The storage humidity shall be up to 95 percent relative.

3.5.8 Operating temperature. - The resolvers shall be capable of giving the performance required herein, as specified, under the testing conditions of and to the requirements of the following:

3.5.8.1 Low temperature test of 4.8.11.

3.5.8.2 High temperature test of 4.8.12.

3.5.8.3 Extreme temperature test of 4.8.13.

3.5.9 Operating humidity. - The resolvers shall be capable of giving the performance required herein with no damage or other evidence of rust, blistering, varnish condensates or corrosion under the testing conditions of and to the requirements of 4.8.10.

3.5.10 Acceleration, operating condition. - The resolvers shall be capable of withstanding the acceleration requirements of MIL-E-5272, procedure II, 15 g in any direction, while in an operating condition; and during and thereafter shall show no damage.

3.5.11 Pressure, operating condition. - Resolvers shall be capable of withstanding sustained atmospheric pressures from 4.0 to 60.0 pounds per square inch absolute, where the rate of change of such pressures does not exceed +3 psi per second or -5 psi per second, and in combination with the temperature and humidity extremes specified in 3.5.8 and 3.5.9 respectively while in an operating condition; and during and thereafter shall show no damage.

3.6 Clean-up. - Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.7 Workmanship. - The resolvers, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of the specification. Particular care and attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of windings, marking of parts and assemblies, and freedom of parts from burrs and sharp edges. All safety wiring shall be in accordance with MS 33540.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Samples

4.2.1 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of the resolvers shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for preproduction tests at an activity designated in the contract or by the Bureau of Naval Weapons for purposes of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of the tests on the preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.2.2 Periodic production samples. - A random sample of a number of units to be specified by the Government inspector will be selected by the Government inspector for submission to a Bureau of Naval Weapons assigned activity to determine compliance of the samples with the requirements of the contract, drawings and specifications. The resolvers shall be considered satisfactory if they meet all specifications and pertinent drawing requirements and withstand the tests specified in 4.7 and 4.8 inclusive.

4.3 Inspections

4.3.1 Government inspection. - The Government inspector shall make such inspections as are necessary to determine that components, assemblies and sub-assemblies are in accordance with the requirements of the contract, pertinent drawings and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

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4.3.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of MIL-Q-21549.

4.4 Classification of tests. - The inspection and testing of the resolvers shall be classified as follows.

4.4.1 Acceptance tests. - These tests are to be accomplished on each resolver being submitted for acceptance under contract Acceptance tests shall be either performed by the manufacturer and witnessed by the Government inspector, or these tests may be conducted by the Government inspector. Acceptance tests are described in 4.7. The acceptance test schedule is listed in table II.

4.4.2 Preproduction and periodic production tests. - Preproduction and periodic production tests are those which are accomplished on samples selected as specified in 4.2.1 and 4.2.2, which are representative of the production of the resolvers, after the award of the contract to determine that the production resolver meets the requirements of this specification as detailed in 4.8 herein. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with the requirements stated herein will result in the rejection of the lot or the cessation of the production, as determined by the procuing activity. The preproduction and periodic production test schedule is listed in table III

4.4.3 Tests after delivery. - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all of the tests specified herein as deemed necessary after delivery of the resolvers. Any defects discovered during or after these tests due to the contractor's failure to comply with the drawings or specifications of this equipment shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.5 Testing practices. - Prior to performing any resolver tests, test personnel shall become familar with the test considerations and theory delineated in OD 23404, and specific test procedures shall be conducted in accordance with the general practices described therein.

4.6 Test equipment. - The equipment and circuits required to conduct resolver tests shall be as described in the appendix of this specification.

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Table II. Acceptance Testing Schedule (100 percent inspection).

Test	Requirement paragraph	Test paragraph
Visual and dimensional	3.22 through 3.2.10 and 3.2.13	4.7.4.1
Continuity	3.4.24	4.7.4.3.5
Friction torque	3.4.23	4.7.4.3.6
Resolver zero marking	3.4.5	4.7.4.2.5
Total null voltage (temp of 4.7.2.1)	3.4.11	4.7.4.2.8
Transformation ratio (K_e)	3.4.14	4.7.4.3.1
Phase shift	3.4.15	4.7.4.3.2
Angular error (temp of 4.7.2.1)	3.4.16	4.7.4.3.3
High potential (leakage resistance)	3.4.8	4.7.4.2.6
Insulation resistance	3.4.9	4.7.4.2.7
Impedance	3.4.4	4.7.4.2.4
Impedance balance	3.4.17	4.7.4.3.4
Compensators (windings and resistors)	3.4.8.3 and 3.4.18 through 3.4.21 inclusive	4.7.4.4

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Table II. Preproduction and Periodic Production Test Schedule.

Test	Requirement paragraph	Test paragraph
All acceptance tests (except high potential)	(See table II)	(See table II)
Zero shift of null spacing at 45 degrees	3.4.12 and 3.4.13	4.8.15
Phase shift variation due to voltage	3.4.13.1	4.8.16
Radio interference	3.4.22	4.8.7
Temperature rise	3.4.25	4.8.14
Low temperature operating tests	3.5.8.1	4.8.11
High potential (leakage resistance)	3.4.8	4.8.17
~~High temperature operating test~~	3.5.8.2	4.8.12
~~Extreme temperature tests~~	3.5.8.3	4.8.13
Vibration tests	3.5.1	4.8.2
Shock tests	3.5.3	4.8.9
Operating life tests	3.3	4.8.3
Acceleration	3.5.10	4.8.19
Humidity	3.5.9	4.8.10
Pressure	3.5.11	4.8.20
Salt water (corrosion) tests	3.5.6	4.8.5
Fungus	3.5.5	4.8.6
Total null voltage (-55 to +85 degrees C)	3.4.11	4.8.18.1
Harmonic distortion	3.4.7	4.8.18.2
Phase shift versus rotation	3.4.13.2	4.8.21
Friction torque	3.4.23	4.8.18.3
Phase rotation	3.4.6	4.8.18.4
Internal capacity	3.4.10	4.8.22

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4.7 Acceptance tests

4.7.1 Testing. - The contractor or Government inspector shall conduct acceptance tests as specified in 4.4.1 to assure that the resolvers are in compliance with the requirements stated herein.

4.7.2 Test conditions. - Unless otherwise specified, the resolvers shall be subjected to acceptance tests under the following conditions:

4.7.2.1 Temperature: Room ambient, 22 ± 2 degrees C, or as otherwise specified in the applicable detail specification.

4.7.2.2 Altitude: Normal ground atmospheric pressure.

4.7.2.3 Vibration: None.

4.7.2.4 Humidity: Room ambient to 95 percent relative maximum.

4.7.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment and facilities and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity; and if definite requirements are not specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.7.4 Test procedure. - The resolvers shall be subjected to the following tests:

4.7.4.1 General inspection. - A comprehensive inspection shall be made to determine the extent of compliance with the requirements of this specification and shall include the determination of the following:

4.7.4.1.1 Size, weight, appearance, and markings.

4.7.4.1.2 Installation difficulties, if any.

4.7.4.1.3 Wiring, cabling, soldering, and color coding.

4.7.4.1.4 Accessibility and ease of servicing and maintenance (if applicable).

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4.7.4.1.5 Adequacy of parts, compliance with specific ratings, and conformance to applicable specifications, and especially the conformance of the mounting surfaces to drawing specifications.

4.7.4.1.6 Errors, if any, in instruction sheets and parts list.

4.7.4.1.7 Suitability of slip-ring and brush assembly (if applicable).

4.7.4.2 Electrical tests. - The resolvers shall be tested in accordance with the requirements of section 3 and in compliance with the test procedures described therein, or in section 4, or failing delineation in either section, then in compliance with the best commercial practice for conformance to the electrical and performance characteristics specified herein as follows:

4.7.4.2.1 Input voltage to 3.4.1.

4.7.4.2.2 Test voltage to 3.4.3.

4.7.4.2.3 Input voltage harmonics to 3.4.2.

4.7.4.2.4 Impedance to 3.4.4.

4.7.4.2.5 Resolver zero marking to 3.4.5.

4.7.4.2.6 Leakage resistance (high potential) to 3.4.8 at 80 percent of the voltages specified therein (3.4.8.4).

4.7.4.2.7 Insulation resistance to 3.4.9.

4.7.4.2.8 Total null voltage to 3.4.11. The readings shall be taken at each angle specified in the angular accuracy test. The test connections shall be as described in figures A7 and A7A for the RX and the RC units respectively.

4.7.4.3 Resolver characteristic tests. - The resolvers shall be tested in accordance with the requirements of section 3, and in compliance with the test procedures described therein, or in section 4, or, failing delineation in either section, then in compliance with the best commercial practice for conformance to the characteristics and performance requirements specified herein as follows:

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4.7.4.3.1 Transformation ratio. - The transformation ratio shall be determined by the following tests to the requirements of 3.4.14.

4.7.4.3.1.1 Test connections. - For resolvers whose transformation ratio is greater than unity, the test connection shall be as described in figure A2 or A2A of appendix. For resolvers whose transformation ratio is less than unity, the test connections shall be as described in figure A3 of appendix.

4.7.4.3.1.2 Test procedures. - The resolvers shall be mounted on the angular accuracy index stand and set at electrical zero. Either the rotor or the stator shall be energized as specified in the detail specification. The test connection circuit shall be energized with the test voltage and frequency specified in the detail specification for the resolver under test. The ratio transformer shall then be adjusted to show a null voltage on the detecting instrument (null detector). K_i (6.4.2.5.1) shall be read or calculated from the dial settings of the ratio transformer, and if applicable, the tap setting of the step-up transformer. K_e (6.4.2.5.2) shall then be derived from the K_i value.

4.7.4.3.2 Phase shift. - The phase shift shall be measured with the resolver essentially unloaded and at the point of maximum coupling as determined under the tests of 4.7.4.3.1 by a phase angle measuring instrument having a minimum sensitivity of 0.2 degrees and to the requirements of 3.4.15.

4.7.3.3.3 Angular error. - The angular error shall be tested to the requirements of 3.4.16 as follows: The resolver shall be mounted on the angular accuracy index stand and its electrical zero shall be made to coincide with the stand zero. Readings will be taken every 5 degrees unless otherwise specified in the detail specification, and the angular departure from the true angle shall be recorded. Each unit shall be tested twice, once as an RX and once as an RC, unless otherwise specified in the detail specification. The test connections shall be as described in figures A7 and A7A for the RX and RC units respectively.

4.7.4.3.4 Impedance balance. - The impedance balance of the resolvers shall be tested to the requirements of 3.4.17 as follows:

4.7.4.3.4.1 Secondary tests, resistive. - The resolver shall be mounted on the angular accuracy index stand and its electrical zero, determined as described in 4.7.4.2.5, shall be made to coincide with the stand zero. The resolver shall be connected as shown in figure A4, and the bridge and resolver shall be set to an electrical angle of plus or minus 45 degrees. Two resistors, Rzs, shall be connected as shown in

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the figure. The resistance of each of the resistors shall be 10 times the nominal output short circuit impedance ± 5 percent, and the resistances shall be equal within ± 1 percent. The resolver shall be re-nulled and the angular shift in seconds shall be measured and recorded. The resistors shall be interchanged across the windings of the resolver and a new null shall be found, measured, and recorded. Z-sec shall be calculated as equal to 0.5 times the algebraic sum of the two recorded angular shifts in seconds.

4.7.4.3.4.2 Secondary tests, reactive. - The resolvers shall be tested as described in 4.7.4.3.4.1 except that two capacitors, Czs, shall replace the two resistors, Rzs. The reactance of each of the capacitors shall be 10 times the nominal output short-circuit impedance ± 5 percent at test frequency, and the reactances shall be equal within ± 1 percent.

4.7.4.3.4.3 Primary tests, resistive. - The resolvers shall be tested as described in 4.7.4.3.4.1 except that the circuit shown in figure A5 shall be used and two resistors, Rzp, shall replace the resistors, Rzs, specified therein. The resistance of each of the resistors shall be 0.1 times the nominal input open-circuit impedance ± 5 percent, and the resistances shall be equal within ± 1 percent

4.7.4.3.4.4 Primary tests, reactive. - The resolvers shall be tested as specified in 4.7.4.3.4.3 except that two capacitors, Czp, shall replace the resistors, Rzp, specified therein. The reactance of each of the capacitors shall be 0.1 times the nominal input open-circuit impedance ± 5 percent at test frequency, and the reactances shall be equal within ± 1 percent.

4.7.4.3.5 Slip-ring brush continuity. - The slip ring-brush continuity shall be tested to the requirements of 3.4.24 as follows: The shaft shall be rotated for several revolutions. With a Wheatstone bridge of the required accuracy across the terminals of each rotor winding, the continuity through the brushes and slip rings shall be determined. Test current shall be limited to less than 10 milliamperes. The test shall be made while the rotor is being slowly turned at a speed of not more than 5 rpm for not less than 2 revolutions in each direction.

4.7.4.3.6 Friction torque. - The friction torque of each resolver shall be tested to the requirements of 3.4.23 as follows: A pendulum which shall exert a torque of not more than the value specified in the detail specification shall be mounted on the resolver shaft. The stator shall then be rotated at a rate not greater than 6 rpm with the shaft in a horizontal position. At least 2 revolutions of the stator shall be made in each direction, ccw and cw. The specified torque value

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shall be considered exceeded if the pendulum turns one revolution.

4.7.4.4 Compensating windings and resistors. - The compensating windings or resistors shall conform to all of the requirements, as applicable herein, for leakage resistance (3.4.8.3) and insulation resistance and, in addition, shall be tested for null voltage, winding phase and equality of null voltage as follows:

4.7.4.4.1 Null voltage. - The resolvers shall be connected in the circuits of figures 7 and 8 respectively, and the readings of the null meter shall conform to the requirements of 3.4.18.

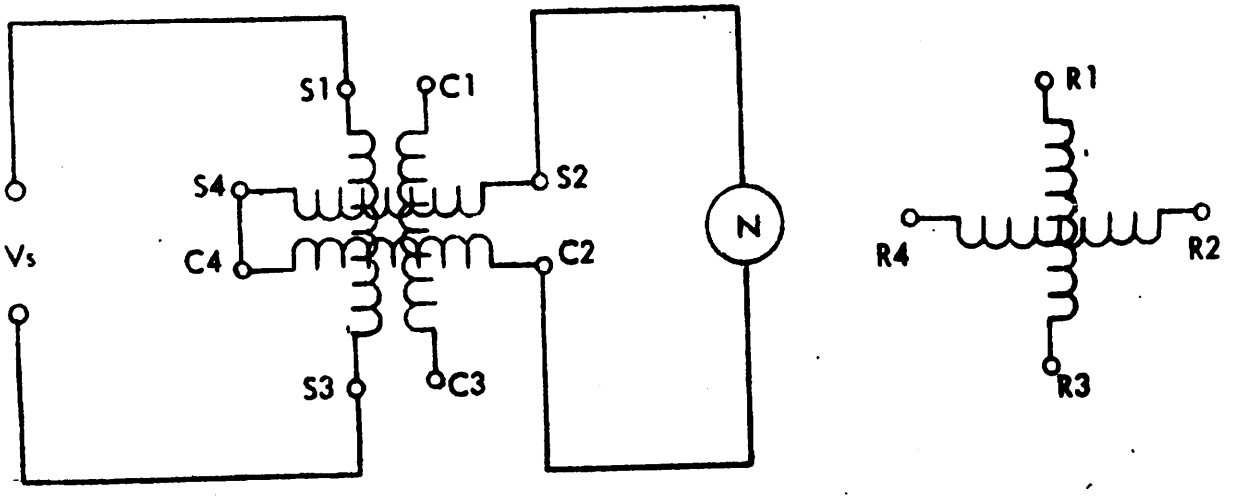


Figure 7

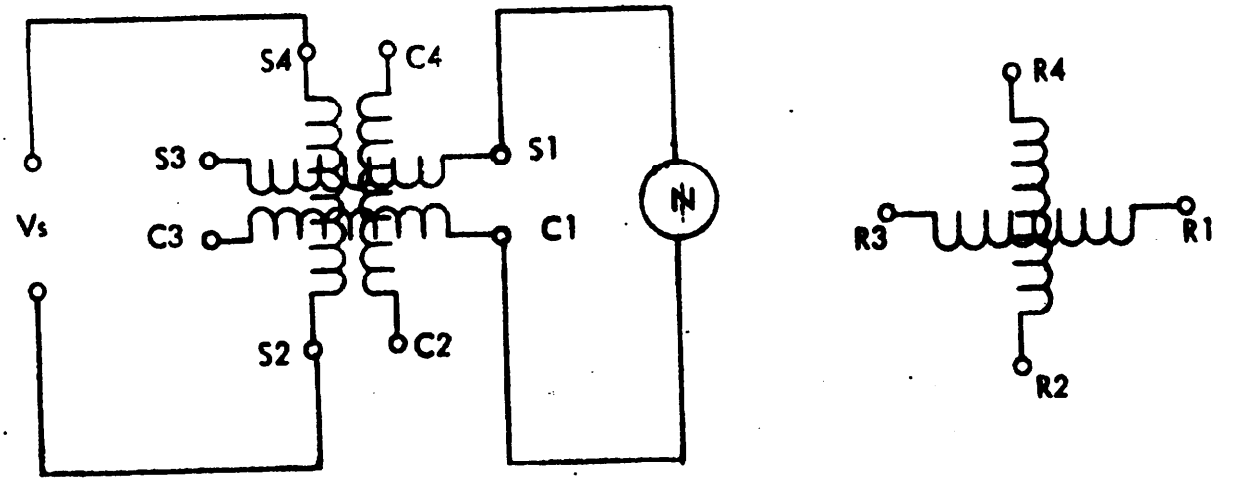


Figure 8

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4.7.4.4.2 Winding phase. - The resolver shall be connected in the circuits of figures 9 and 10 respectively, and the readings of the voltmeter shall conform to the requirements of 3.4.19.

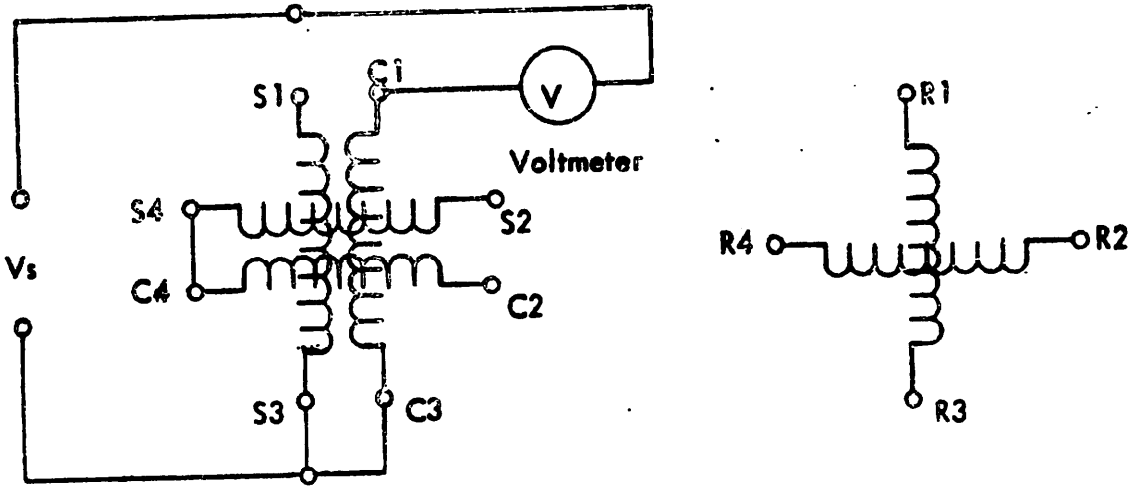


Figure 9

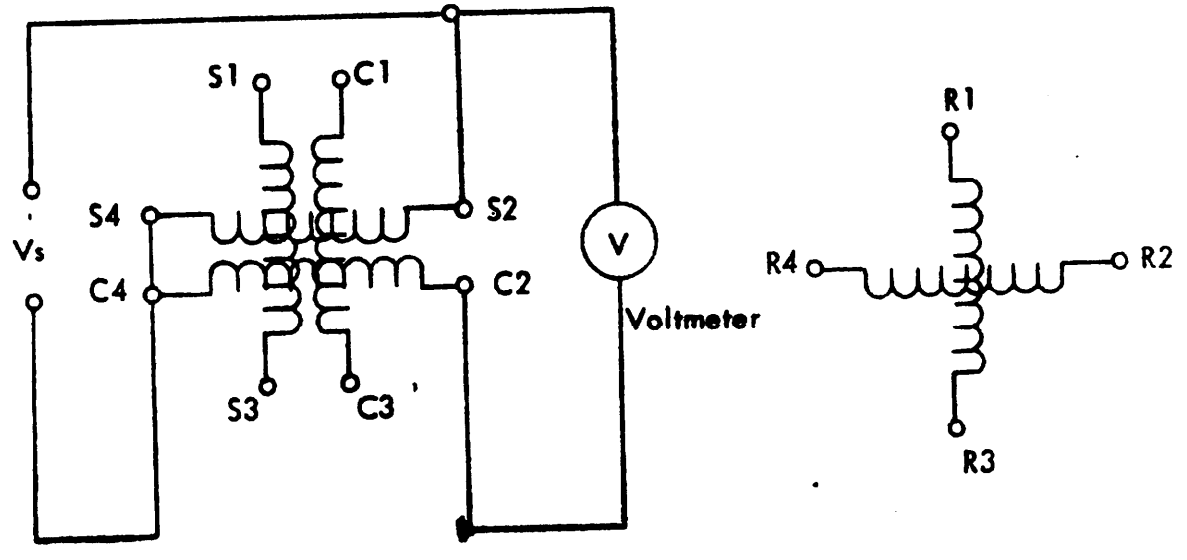


Figure 10

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4.7.4.4.3 Equality of null voltage. - The resolver shall be connected in the circuit of figure 11 and the reading of the null meter shall conform to the requirements of 3.4.20.

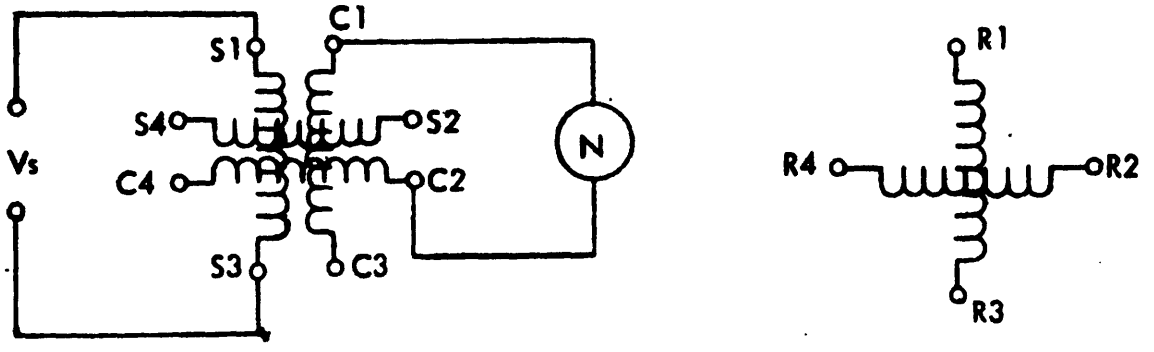


Figure 11

4.7.4.4.4 Compensating resistor. - When applicable, the compensating resistors shall be checked for the proper value, material and terminal markings as specified in 3.4.21.

4.8 Preproduction and periodic production tests. - As specified in the contract or order, the contractor or Government inspector or the procuring activity shall conduct preproduction and periodic production tests as specified in 4.4.2 and tests after delivery as specified in 4.4.3 to assure that the resolvers are in compliance with the requirements specified herein.

4.8.1 Operating conditions for environmental tests. - When specified herein, the resolver shall be mounted in the environmental test fixture and energized with test voltage. The resolver outputs shall be loaded with two resistors, each of which is 10 times the nominal output short-circuit impedance ± 5 percent, and which are equal within ± 1 percent. The resolver shall be set and locked at 45 degrees electrical angle and monitored by means of a resolver bridge and null meter to indicate angular shift.

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4.8.2 Vibration, operating condition. - The resolvers shall be rigidly mounted in a suitable vibration fixture as specified in 4.6.1 and shall be subjected to vibration tests in accordance with 3.5.1. During the test the angular deviation shall not exceed the values specified in the detail specification. Each time the fixture is remounted to vibrate on another axis, contractor may reset the angle of the resolver. The resolver shall, after the tests, meet the requirements of 4.7.

4.8.3 Accelerated life tests. - The resolvers shall be subjected to the following accelerated life tests. The resolvers shall be operated continuously at full rate load (a resistive load equal to 10 times nominal output short circuit impedance) at the upper range of the input voltage and at rated frequency. The operation shall continue for 1500 hours at a speed of 1150 ± 50 rpm (or as otherwise specified in the detail specification) in an ambient temperature of 71 degrees C minimum and a humidity of at least 50 percent relative; and thereafter shall meet all of the performance requirements of 4.7.

4.8.4 Vibration, shipping condition. - The resolver, rigidly mounted in a suitable vibration fixture, shall be subjected to vibration tests so as to meet the requirements specified in 3.5.2; and thereafter shall meet all the requirements of 4.7.

4.8.5 Salt water (corrosion) tests. - The resolvers shall be exposed to the salt water test so as to meet the requirements specified in 3.5.6; and thereafter shall meet the requirements of 4.7.

4.8.6 Tropical fungus tests. - The resolvers shall be subjected to the fungus tests so as to meet the requirements specified in 3.5.5, and thereafter shall meet the requirements of 4.7.

4.8.7 Radio interference tests. - The resolvers shall be energized and the rotor rotated at 1150 ± 50 rpm. The rod antenna of a noise meter in conformance to the requirements of MIL-I-16910 shall be placed within 3.0 ± 0.3 feet of the resolver brushes and the noise level shall be measured and recorded at no less than 3 frequencies in each of the bands between 0.15 and 150 mc. The resolvers shall meet the requirements of 3.4.22. During the tests the resolvers shall be energized as follows:

4.8.7.1 Rotor energized units. - The primary shall be R1-R3 and shall be energized with 0.707 of the maximum input voltage ± 5 percent. The secondaries (S1-S3 and S2-S4) shall be loaded with 10K ohm resistors, terminal R2 shall be

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shorted to R1, and terminal R4 shall be shorted to R3.

4.8.7.2 Stator energized units. - The primary shall be S1-S3 and shall be energized with 0.707 of the maximum input voltage ± 5 percent. The secondaries R1-R3 and R2-R4 shall each be loaded with 10,000 ohm resistors. The terminal S2 shall be shorted to S1 and the terminal S4 shall be shorted to S3.

4.8.8 Shock, shipping condition. - The resolvers, packed and packaged as specified in section 5 herein, shall be tested in accordance with the shock conditions specified in 3.5.4 and thereafter meet all the requirements of 4.7.

4.8.9 Shock, operating condition. - The resolvers shall be mounted in the test fixture as specified in 4.8.1 and tested in accordance with 3.5.3. Variation in electrical angle during test shall meet the requirements of the detail specification.

4.8.10 Humidity, operating condition. - The resolvers shall be prepared for test as specified in 4.8.1 and shall be subjected to the humidity test, method 103 of MIL-STD-202, test condition A. During the test the angular deviations shall not exceed the values specified in the detail specification. Thereafter the resolvers shall be subjected to and shall meet all the following tests.

4.8.10.1 Friction torque of 4.7.4.3.6.

4.8.10.2 Transformation ratio of 4.7.4.3.1.

4.8.10.3 Angular error to 4.7.4.3.3.

4.8.11 Low-temperature tests. - The resolvers, mounted in the test fixture as specified in 4.8.1 and using applicable the test connections of A6 and A6A, shall be subjected to a temperature of -55 ± 5 degrees C for 4 hours so as to meet requirements of 3.5.8.1 The variations in electrical angle, as interpreted from bridge null voltage, shall not exceed the requirements specified in the detail specification. Readings shall be taken every 30 minutes during the 4 hour period that the chamber is at stable temperature.

4.8.12 High-temperature tests. - The resolvers, mounted in the test fixture as specified in 4.8.1 and using the applicable test connections of A6 and A6A, shall be subjected to the temperature of +85 ± 5 degrees C for 4 hours so as to meet the requirements of 3.5.8.2. The variations in electrical angle, as interpreted from

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bridge null voltage shall not exceed the requirements specified in the detail specification. Readings shall be taken every 30 minutes during the 4 hour period that chamber is at stable temperature.

4.8.13 Extreme temperature tests.- The resolvers shall be tested by operation at 125 degrees C to the requirements of 3.5.8.3 and shall be tested as follows after 4 hours at thermal equilibrium with the referenced temperature:

4.8.13.1 High potential tests of 4.8.17 to the requirements of 3.4.8.

4.8.13.2 Insulation resistance tests of 4.7.4.2.7 to the requirements of 3.4.9.

4.8.14 Temperature rise.- The resolvers shall be tested for temperature rise for conformance to the requirements of 3.4.25 as follows:

4.8.14.1 The resolvers shall be unmounted and resting on a heat insulating surface in a space which is free from drafts (air velocity less than 1 ft per sec) and in an ambient temperature of 22.0 ± 3.0 degrees C.

4.8.14.2 The resolver shall be connected in the circuit of figure 12 and test voltage and frequency of the detail specification shall be applied as in figure 12.

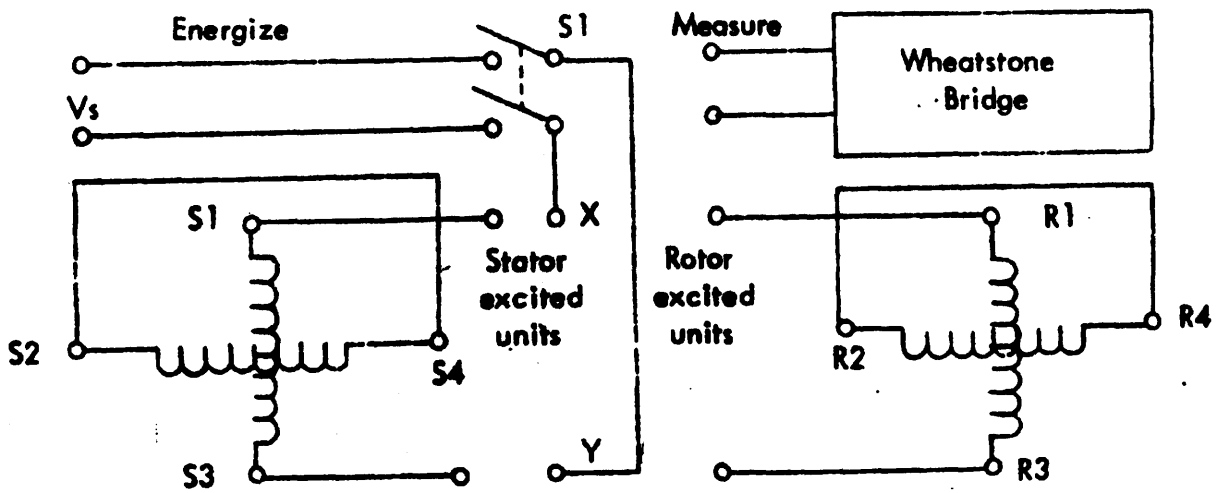


Figure 12

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4.8.14.3 The switch, S1, shall be set to the measure position, and the resistance of the winding, Rc, connected to "X-Y", shall be determined by means of a Wheatstone bridge. The ambient temperature, Tc, at a distance of 6 ± 1 inch from the unit, shall be measured at the same time as the resistance measurement is made to an accuracy of ± 1.0 degree C and shall be recorded.

NOTE

Current applied to the resolver windings by the Wheatstone bridge shall be limited to not more than 10 milliamperes.

4.8.14.4 The switch, S1, shall be set to the energize position, and the unit shall be excited for at least 5 hours. The ambient temperature at a distance of 6 ± 1 inch from the unit shall be measured to an accuracy of ± 1.0 degree C for every 60 minutes ± 30 seconds from the time of excitation.

4.8.14.5 At the conclusion of the excitation period, the switch, S1, shall be set to measure, and as many resistance readings as possible shall be obtained within the first 100 seconds of cooling. The time each reading, starting at the time of resetting of S1 as T zero, shall be recorded. A graph of resistance versus time shall be drawn and the curve extrapolated to cross the T zero axis; the value of resistance at this point shall be taken as Rh.

4.8.14.6 The temperature rise shall be calculated from the following equation:

$$Tr = (234.5 + Tc)\left(\frac{Rb}{Rc} - 1\right) - tc$$

Where the symbols are defined as follows:

4.8.14.6.1 Tr = Temperature rise.

4.8.14.6.2 Rc is the resistance of the winding at the beginning of the test as measured in 4.8.14.3.

4.8.14.6.3 Tc is the initial ambient temperature in degrees C as measured in 4.8.14.3.

4.8.14.6.4 The temperature tc is the average of all ambient temperatures (as measured in 4.8.14.4), minus Tc.

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4.8.14.6.5 Rh is the resistance of the winding at the end of the test as determined from the graph of 4.8.14.5.

4.8.15 Zero shift of null spacing at 45 degrees. - The resolvers shall be tested for zero shift of null spacing at 45 degrees due to voltage and frequency variation as follows.

4.8.15.1 Variation with voltage. - The resolver shall be zeroed on the angular accuracy index stand and the stand shall be rotated to an electrical angle of 45 degrees. Primary voltage shall be varied as specified in 3.4.12, and the angular deviation from 45 degrees shall be recorded.

4.8.15.2 Variation with frequency. - The tests of 4.8.15.1 shall be repeated under the conditions of 3.4.13, and the angular deviation from 45 degrees shall be recorded.

4.8.16 Phase shift variation due to voltage. - The resolvers shall be tested for phase shift due to voltage to the requirements of 3.4.13.1, and the phase shift variation due to voltage shall be measured and recorded.

4.8.17 High potential (leakage resistance). - The resolvers shall be tested for leakage resistance (high potential) as specified in 3.4.8 and in accordance with the requirements thereof.

4.8.18 Characteristic tests. - The following characteristic tests shall be performed.

4.8.18.1 Total null voltage at -55 to +85 degrees C to 3.4.11 in accordance with 4.7.4.2.8.

4.8.18.2 Harmonic distortion to 3.4.7.

4.8.18.3 Friction torque at -55 to +85 degrees C to 3.4.23 in accordance with paragraph 4.7.4.3.6.

4.8.18.4 Phase rotation to 3.4.6.

4.8.19 Acceleration, operating condition test. - The resolver shall be mounted to a suitable acceleration fixture (centrifuge) and prepared for test as

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described in 4.8.1, except that resolver outputs shall not be monitored. The resolver shall then be tested as specified in 3.5.10, and thereafter shall meet all performance requirements specified in 4.7.

4.8.20 Pressure, operating condition test. - The resolver shall be placed in a pressure chamber, shall be subjected to pressure tests so as to meet the requirements of 3.5.11 and thereafter shall meet all the performance requirements of 4.7.

4.8.21 Phase shift variation tests. - The resolver shall be mounted on the angular accuracy index stand and its electrical zero shall be made to coincide with the stand zero. The phase shift measurement shall be from the (1-3) primary winding to the (2-4) secondary winding. Readings shall be taken every 5 degrees of rotation, unless otherwise specified in the detail specification, and the phase angle variation shall be measured and recorded during the tests of the secondary only and shall meet the requirements specified in 3.4.13.2. The phase shift shall not be recorded or measured at zero and 180 degrees.

4.8.22 Internal capacity test. - The internal capacities of the resolver shall be measured and the equivalent capacities shall be determined in accordance with the following equations. Symbols are defined in section 6.

$$C_{FR} = \frac{C_{1R}}{2 - \frac{C_{1R}}{2C_{2R}}}$$

$$C_{WR} = C_{2R} - \frac{(C_{1R})\ (C_{2R})}{4\ (C_{2R}) - (C_{1R})}$$

$$C_{WS} = C_{2S} - \frac{(C_{2S})\ (C_{1S})}{4\ (C_{2S}) - (C_{1S})}$$

4.8.23 Storage life test. - The ability of the resolvers to meet the operating life test, plus the contractor's certification that the unit will meet the storage life requirements specified herein, when packed and packaged in accordance with level A (military pack) requirements, will be considered as satisfying the storage life tests.

4.9 Criteria for rejection. - Failure of the resolvers to meet any of the requirements and tests of this specification shall be considered sufficient cause for re-

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jection of individual resolvers or of the entire lot, as determined by the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - The resolvers shall be preserved and packaged in accordance with one of the levels outlined below, as specified in the contract, order or purchase requisition.

5.1.1 Level A (military package). - Level A packing is that preservation and packaging method that will afford adequate protection against corrosion, deterioration, and physical damage during shipment, handling, indeterminate storage and world-wide distribution under military conditions and consists of the following:

5.1.1.1 Cleaning. - Cleaning for level A packaging shall be selected in accordance with MIL-P-116, with consideration given to the intended use and the defined level, and shall be adequate to insure compliance with 3.5.7.

5.1.1.2 Drying. - Drying for level A preservation and packaging shall be selected in accordance with MIL-P-116, with consideration given to the intended use and the defined level, and shall be adequate to insure compliance with 3.5.7.

5.1.1.3 Preservation. - The preservation used (within the requirements of the preservation application criteria of MIL-P-116) shall be selected in accordance with MIL-P-116, with consideration given to the intended use and the defined level, and shall be adequate to insure that the requirements of 3.5.7 are met.

5.1.1.4 Unit protection. - The resolvers shall individually be afforded unit protection in accordance with method IA-7 of MIL-P-116.

5.1.1.5 Intermediate packaging. - Ten unit packs of 5.1.1.4, or as determined by the contractual document, shall be placed in an intermediate container selected in accordance with, and to the requirements of, MIL-P-16298 for level A packaging requirements.

5.1.2 Level B (limited military package). - Level B preservation and packaging is that which will afford adequate protection against damage during multiple domestic shipments, handling, and covered storage, and consists of the following:

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5.1.2.1 Cleaning. - Cleaning for level B preservation shall be selected in accordance with MIL-P-116, with consideration given to the intended use and the defined level, and shall be adequate to insure compliance with 3.5.7 as modified by 6.2 and 6.3.

5.1.2.2 Drying. - Drying for level B packaging shall be selected in accordance with MIL-P-116, with consideration given to the intended use and the defined level, and shall be adequate to insure compliance with 3.5.7 as modified by 6.2 and 6.3.

5.1.2.3 Preservation. - The preservative used (within the requirements of the preservative application of MIL-P-116) shall be selected in accordance with MIL-P-116, with consideration given to the intended use and the defined level, and shall be adequate to insure that the requirements of 3.5.7 as modified by 6.2 and 6.3 are met.

5.1.2.4 Unit protection. - The resolvers shall individually be afforded unit protection in accordance with method IA-5 of MIL-P-116.

5.1.2.5 Intermediate packaging. - Ten unit packs of 5.1.2.4, or as determined by the contractual document, shall be placed in an intermediate container selected in accordance with, and to the requirements of, MIL-P-16298 for level B packaging requirements.

5.1.3 Level C (minimum military package). - Level C packaging is that which will afford protection against damage during the direct shipment from the supply source to the first receiving activity for immediate use. This level, in general, will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level. It shall consist of the following:

5.1.3.1 Cleaning. - Cleaning for level C packaging shall be selected in accordance with MIL-P-116, with consideration given to the intended use and the defined level.

5.1.3.2 Drying. - Drying for level C packaging shall be selected in accordance with MIL-P-116, with consideration given to the intended use and the defined level.

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5.1.3.3 Preservation. - The preservative used (within the requirements of the preservation application criteria of MIL-P-116) shall be selected in accordance with MIL-P-116, with consideration given to the intended use and the defined level. In the event no preservative is used, but both cleaning and drying are accomplished, all bearings shall be relubricated in accordance with the bearing manufacturer's recommendations.

5.1.3.4 Unit protection. - The resolvers shall individually be afforded unit protection in accordance with method IA-5 of MIL-P-116.

5.1.3.5 Intermediate packaging. - The unit packs of 5.1.3.4, in the number specified in the contractual document, shall be placed in an intermediate container selected in accordance with, and to the requirements of, MIL-P-16298 for level C packing requirements.

5.2 Packing. - Exterior packing shall be as follows:

5.2.1 Level A packing. - The exterior shipping container for level A packing shall conform to the requirements of MIL-P-16298 for the defined level.

5.2.2 Level B packing. - The exterior shipping container for level B packing shall conform to the requirements of MIL-P-16298 for the defined level.

5.2.3 Level C packing. - For level C packing, intermediate packs shall be protected from damage for direct domestic shipment from the supply source to the first receiving activity for immediate use. Level C, in general, will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

5.3 Marking. - Labeling and marking of all shipping containers, packs and unit containers shall be in accordance with MIL-STD-129. Additional information as required by procuring activity for unit pack labeling under level C requirements may also be applied to the unit packs.

5.4 Certification. - For level A packaging and packing, the contractor shall enclose certification that the package will meet the requirements of 3.5.7 of this specification.

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6. NOTES

6.1 Ordering data. - Procurement documents for a given purchase order should specify the title, number, and date of this specification, the applicable BuWeps drawing and the detail specification for the resolver ordered, and the level of packaging and packing required, and should specify as applicable those items requiring contract delineation as listed in table IV. The number of unit packs to an intermediate package should also be specified.

Table IV. Items Requiring Contract Delineation

Item	Paragraph
Preproduction samples	3.2.1
Materials	3.2.2
Use of magnesium	3.2.2.4
Fungistatic materials	3.2.5.1
Inspection requirements	4.1
Preproduction samples	4.2.1
Periodic inspection samples	4.2.2
Tests after delivery	4.4.3
Packing and packaging	5.1

6.2 Level B limitations. - Level B preservation and packing is intended to provide economical but limited protection (to a maximum of one year under the conditions of 3.5.7 herein) and should be specified only when it is determined to be appropriate.

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6.3 Storage surveillance. - Items preserved and packaged in accordance with level B requirements must be inspected to determine condition when not used within the time period specified. Items not used within this time period must either be preserved or repackaged in accordance with level B requirements of this specification, or level A requirements if storage beyond ad additional year is anticipated.

6.4 Definitions

6.4.1 Terms. - The following definitions are applicable to this specification for the terms used herein.

6.4.1.1 Angular error. - The angular error is the difference between the resolver mechanical angle and its true electrical angle.

6.4.1.2 Angular position. - The angular position of the resolver is the angular displacement of its rotor about the axis of the rotor from the resolver's zero position measured as an increasing positive angle in the counter clockwise direction.

6.4.1.3 Compensating winding. - The compensating winding is an additional (stator) winding to adjust the transformation ratio, phase shift and loading.

6.4.1.4 Direction of rotation. - The direction of rotation is specified as viewed from the shaft extension end of the resolvers, or the mounting and in the case where no shaft extension exists, or, in the case of special resolvers, as specified on the detail drawing.

6.4.1.5 Electrical angle theta. - The electrical angle theta is the angle which satisfies the relative magnitudes and polarities of the secondary voltages of an ideal resolver in accordance with the equations of 3.1.1 and 3.1.2 and the vector conventions of 6.4.1.16.

6.4.1.6 Equality of transformation ratio. - Equality of transformation ratio refers to the variation of the transformation ratio of all possible combinations of stator and rotor windings from primary to secondary.

6.4.1.7 Friction torque. - The friction torque is the amount of torque required to overcome the friction of the rotor at a specified temperature and with no power applied to the resolver.

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6.4.1.8 Function error. - Function error is the deviation of the resolver output voltages from those deriving from the equations of 3.1.1 and 3.1.2 and is given as a percentage of the maximum magnitude of the ideal sinusoid.

6.4.1.9 Fundamental null voltage. - The fundamental null voltage is the minimum secondary voltage of the excitation frequency.

6.4.1.10 Impedance balance. - The impedance unbalance in Z-seconds is a measure of the error which results from loading the resolver.

6.4.1.11 Input current. - Input current is the current drawn by the primary when the rated input voltage is applied and the secondaries are unloaded.

6.4.1.12 Input power. - Input power is the power consumed by the primary when the rated input voltage is applied and the secondaries are unloaded.

6.4.1.13 Mode of excitation. - The mode of excitation indicates whether the stator or the rotor winding is energized (primary winding), i.e., rotor excited or stator excited.

6.4.1.14 Nominal. - The word nominal, as used herein, refers to the "design-center" value to which tolerances are applied.

6.4.1.15 Null angle. - The null angle is that angular position of the rotor at which the fundamental frequency component of voltage in phase with the nominal voltage phase of the secondary winding is zero.

6.4.1.16 Phase shift. - Phase shift is the time phase angle of the secondary voltage referred to the primary voltage.

6.4.1.17 Resolver or electrical zero. - Resolver or electrical zero is defined as rotor (inner element) position produced under the following conditions:

6.4.1.17.1 Rotor excited units. - That position of the rotor for which windings (R1-R3 and S2-S4) experience minimum coupling (null) and for which the established vector convention is in accordance with figure 13 and the equations of 3.1.1 for positive (counterclockwise) rotation.

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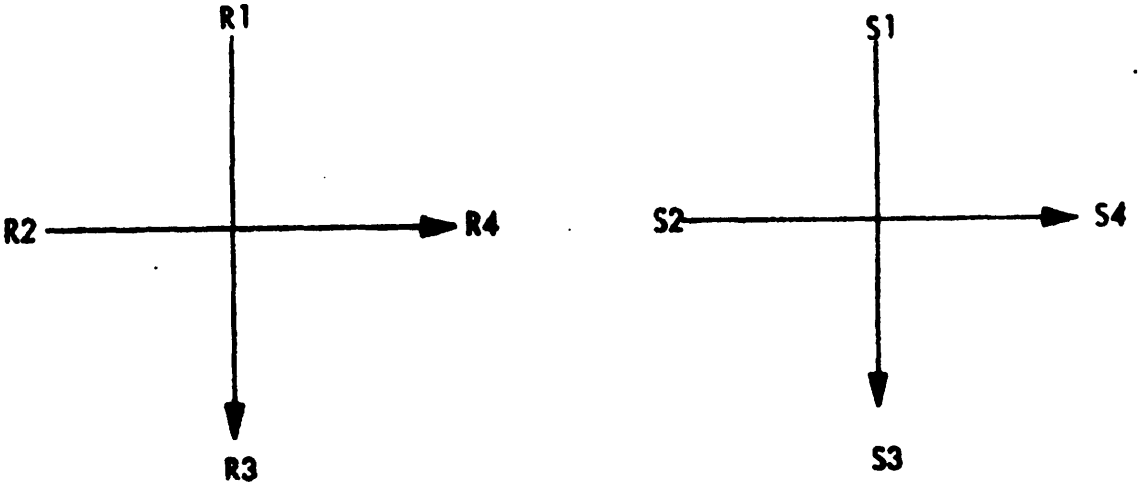


Figure 13

6.4.1.17.2 Stator excited units. - That position of the rotor for which the windings (S1-S3 and R2-R4) experience minimum coupling (null) and for which the established vector convention is in accordance with figure 14 and the equations of 3.1.2 for positive (counterclockwise) rotation.

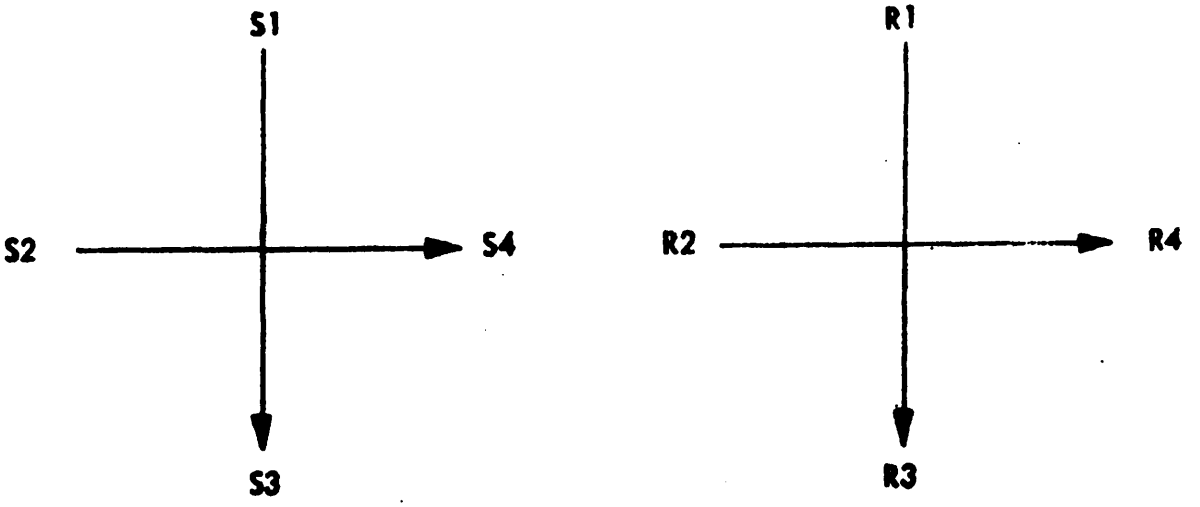


Figure 14

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6.4.1.18 Total null voltage. - The total null voltage is the minimum secondary voltage, including harmonics.

6.4.1.19 Transformation ratio.. - Transformation ratio (K) is the general designation for the ratio of the magnitude of the output voltage at maximum coupling to the magnitude of the input voltage (see 6.4.2.5).

6.4.2 Symbols. - The following symbols are applicable to this specification for the terms used herein:

6.4.2.1 AAIS: Angular accuracy index stand.

6.4.2.2 E or V_S: Input voltage in rms volts.

6.4.2.3 HWA: Harmonic wave analyzer.

6.4.2.4 Internal capacities. - The principal equivalent internal capacities of a resolver are represented in figure 15. These capacities cannot be measured directly; they are computed from measured values as described in 4.8.22.

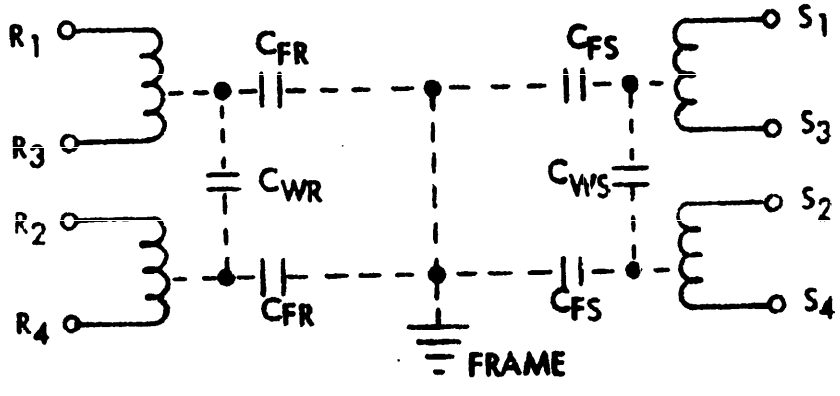


Figure 15. Principal internal capacities

Capacitance symbols are defined as follows:

6.4.2.4.1 C_{FR}: Capacity from rotor winding to frame.

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6.4.2.4.2 C_{WR}: Capacity between rotor windings.

6.4.2.4.3 C_{FS}: Capacity from stator winding to frame.

6.4.2.4.4 C_{WS} Capacity between stator windings.

6.4.2.4.5 C_{1R}. Measurement between rotor winding and frame.

6.4.2.4.6 C_{1S}: Measurement between stator winding and frame.

6.4.2.4.7 C_{2R}: Measurement between rotor windings.

6.4.2.4.8 C_{2S}: Measurement between stator windings.

6.4.2.5 K: General designation for transformation ratio (6.4.1.19). Specific designations are as follows:

6.4.2.5.1 K_i: "In-phase" transformation ratio. The component of the secondary voltage, at maximum coupling, which is in phase with the primary voltage, divided by the primary voltage.

6.4.2.5.2 K_e: "Effective" transformation ratio. K_i divided by the cosine of the nominal phase shift.

6.4.2.6 N; Null meter.

6.4.2.7 RC: Resolver control transformer (4-wire control transformer).

6.4.2.8 RD: Resolver differential (4-wire differential).

6.4.2.9 RX: Resolver transmitter (4-wire transmitter).

6.4.2.10 Vr: Reference voltage in phase with nominal output voltage of resolver at maximum coupling.

6.4.2.11 Vs: Test voltage in accordance with the detail specification.

6.4.2.12 Zco: Compensator impedance, rotor open-circuited.

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6.4.2.13 Z-in: Input impedance.

6.4.2.14 Zro: Rotor impedance, stator open-circuited.

6.4.2.15 Zrs: Rotor impedance, stator short-circuited.

6.4.2.16 Z-sec: Impedance balance unit (6.4.1.10).

6.4.2.17 Zso: Stator impedance, rotor open-circuited.

6.4.2.18 Zss: Stator impedance, rotor short-circuited.

6.4.2.19 Zt: Transmitting impedance

6.4.2.20 θ: Electrical angle (6.4.1.5).

6.4.3 Winding convention. - Whenever a winding is referenced in terms of its enumerated termination letters (e.g., S1-S2), the direction of the instantaneous voltage vector shall be from the first enumerated letters to the second.

6.5 Application and theory. - Additional information regarding recommended application and theory of operation of resolvers covered by this specification is contained in OD 23404.

6.6 Requests for deviation or waiver. - Requests for deviation or waiver from this specification, applicable drawings, specifications, publications, and materials and processes specified herein shall be as outlined in OP 400.

6.7 Oral statements. - Oral statements by any person or persons shall not be allowed in any manner or degree to modify or otherwise affect the requirements of any art of this specification, or any specification, publication or drawing referenced herein.

6.8 Conflicting requirements. - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring activity or appointed agent for interpretation, clarification and resolution (or correction).

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APPENDIX

TEST EQUIPMENT

10. SCOPE

10.1 This appendix covers the requirements for test equipments and circuits which are required to perform the tests described in section 4 of this specification.

20. CIRCUIT DESCRIPTIONS

20.1 Resolver impedance bridge. - The bridge required for measurement of impedance parameters Zro, Zrs, Zso, and Zss shall be of the resonant type, incorporating a resonating decade capacitor (C_D) in the same arm as the resolver under test. The resistive component of the resolver impedance shall be balanced and read on a decade resistor (R_D) placed in a suitable arm, preferably in the series-current branch with the resolver to provide operation as an equal-ratio bridge.

20.1.1 Design. - The impedance bridge may be of the classical configuration shown in Circuit A of figure A1; however, less difficulty may be experienced with stray reactances if the configuration of Circuit B is used.

20.1.2 Electrical supply characteristics. - The impedance bridge shall be capable of energizing the resolver under test at its operating voltage and frequency for open-circuit tests and at its open-circuit current and operating frequency for short-circuit tests.

20.1.3 Impedance ranges and accuracy. - The impedance bridge shall be capable of measuring the resistive and reactive components of resolver impedance over the following ranges. (In those instances where an impedance parameter below the reactive range of the bridge must be measured, plug-in capacitor must be paralleled with the decade capacitor.)

(a) 1 to 100 K ohms resistive and +j18 to +j40 K reactive at 800 cps, with an accuracy of ± 1 percent or 0.1 ohm (whichever is greater) from 1 to 10 K ohms resistive and from j18 to j10 K ohms reactive, and ± 2 percent from 10 K to 100 K ohms resistive and from j10 K to j40 K ohms reactive.

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(b) 1 to 100 K ohms resistive and +j36 to +j80 K ohms reactive at 400 cps, with an accuracy of ± 1 percent or 0.1 ohm (whichever is greater) from 1 to 10 K ohms resistive and from j36 to j10 K ohms reactive, and ± 2 percent from 10 K to 100 K ohms resistive and from j10 K to j80 K ohms reactive.

(c) 1 to 100 K ohms resistive and ±j1.5 to +j10 K ohms reactive at 10 Kc, with an accuracy of ± 3 percent or 0.1 ohm (whichever is greater) from 1 to 10 K ohms resistive and from j1.5 to j10 K ohms reactive. (It is not anticipated that the bridge will be used to measure resistances greater than 10 K ohms at this frequency.)

20.1.4 Bridge component requirements. - The characteristics of the components (in particular the stray reactances) are quite critical in attaining the required bridge accuracy. The following recommendations apply to the components shown in figure A1.

20.1.4.1 Transformer T1. - Transformer T1 is a fully-shielded impedance bridge input transformer. When used in Circuit B of figure A1, the transformer must have an accurate center-tapped secondary; in the case of Circuit A, the center tap is unnecessary. The transformer should meet the following requirements:

(a) Operating voltage: 1 to 100 volts

(b) Operating frequency: 400 cps through 10 kc

(c) Power capability: 15 volt-amperes

(d) Transformation ratio: 1.0 ± 1 percent

(e) Phase shift: Less than 1 degree

(f) Secondary short-circuit impedance: 50 ohms maximum

(g) Primary open-circuit impedance: 25 K ohms minimum

(h) Accuracy of center tap on secondary: 0.01 percent

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(i) Phase balance between two halves of secondary: Equal within 0.1 milliradian

(j) Short-circuit impedance unbalance from either side of secondary to center tap: 0.1 ohm maximum.

(k) Capacity from secondary to shield: 100 picofarads (pf) maximum

(l) Unbalance of secondary to shield: 10 pf maximum

20.1.4.2 Transformer T2. - Transformer T2 is a primary- and case-shielded bridge output transformer which links the bridge to a null meter. The transformer should have the following characteristics:

(a) Operating voltage: 0.001 to 20 volts

(b) Operating frequency: 400 cps to 10 kc

(c) Power capability: 4.0 volt-amperes

(d) Transformation ratio: 4.0 ± 5 percent

(e) Phase shift: Less than 5 degrees

(f) Primary open-circuit impedance: 1000 ohms minimum

(g) Primary short-circuit impedance: 10 ohms maximum

(h) Parallel self-resonance: 10 kc minimum

(i) Capacity from primary to shield: 20 pf maximum

20.1.4.3 Decade capacitor C_D. - The decade capacitor should consist of 5 decades - 10^2, 10^3, 10^4, 10^5, and 10^6 pf / step - accurate within ± 0.5 percent, with minimum residual capacities and capacities to their shields.

20.1.4.4 Decade resistor R_D. - The decade resistor should consist of 6 decades - 10^{-1}, 10^0, 10^1, 10^2, 10^3, 10^4 ohms/step - accurate to within ± 0.1 percent, with minimum stray capacities in parallel with the resistance and from the resistance to the shield.

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20.1.4.5 Shunt resistor R_S. - Resistor R_S is a 1.0-ohm shunt resistor which is used to measure the current flowing through the resolver under test.

20.1.4.6 Resistor R_P. - Resistor R_P, in Circuit A, shall be a 1000-ohm ± 0.1 percent nonreactive precision power resistor capable of dissipating a minimum of 5 watts.

20.1.4.7 Resistors R_r. - Resistors R_r, in Circuit A, shall consist of three 0.1 percent precision resistors of 10, 100, and 1000 ohms capable of a power dissipation of 5 watts minimum.

20.1.4.8 Null meter. - The null meter shall be an instrument of the quadrature rejecting type, and shall have a minimum sensitivity such that the smallest scale division represents a change in resolver angle of not more than 10 percent of the maximum allowable angular error of the resolver under test. The relative phase shift between the signal and reference channels shall remain constant within ± 1 degree as the signal voltage varies from 0.001 to 10 volts.

20.2 Angular accuracy resolver bridge. - The angular accuracy bridge shall be of the symmetrical four-arm resistive type and of a design which will not directly interconnect the resolver windings. Each arm shall have a total resistance of 10 K ohms. The bridge shall be calibrated at the frequency(s) at which it will be used. The inherent error of the bridge shall not exceed 10 percent of the maximum allowable error of the resolver under test, or 2 seconds of arc (whichever is greater), and it shall be calibrated against suitable standards to insure this accuracy. Additional information concerning the bridge is contained in OD 23404.

20.3 Circuit components and equipment. - The test equipment and components shown in figures A2 through A7A shall meet the following requirements:

20.3.1 Null meter. - The null meter (N) shall be as specified in 20.1.4.8.

20.3.2 Transformer T2. - Transformer T2 shall be as specified in 20.1.4.2.

20.3.3 Angular accuracy index stand (AAIS). - The angular accuracy index stand shall have a calibrated accuracy and readability of ± 2 seconds of arc or ± 10 percent of the maximum allowable angular error of the resolver under test, whichever is greater.

20.3.4 Voltmeter (V). - The vacuum tube voltmeter shall have an input impedance greater than 2 megohms, and shall measure from 3 millivolts full scale to 300 volts full scale, with an accuracy of ± 2 percent of full scale.

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20.3.5 Transformer T3. - Transformer T3 shall be a precision ratio transformer. The accuracy of the indicated ratio shall be at least ± 0.001 percent + 0.0006 percent/the ratio. .

20.3.6 Transformer T4. - Transformer T4 shall be a precision step-up transformer with 100-, 200-, and 300-percent taps. The taps shall be accurate within ± 0.01 percent. The phase shift into the load shall be less than 20 minutes.

20.3.7 Transformer T5. - Transformer T5 is a primary- and case-shielded isolation transformer which connects the null meter to a resistive type resolver bridge. The transformer should have the following characteristics:

(a) Operating voltage: 0.001 to 1.0 volt

(b) Operating frequency: 400 cps through 10 Kc

(c) Transformation ratio: 1:0 ± 5 percent from 0.001 to 1.0 volt at 400 cps through 10 Kc

(d) Phase shift: 1 degree maximum

(e) Primary open-circuit impedance: 25 K ohms minimum from 400 cps to 10 Kc at a voltage level of 0.001 volt

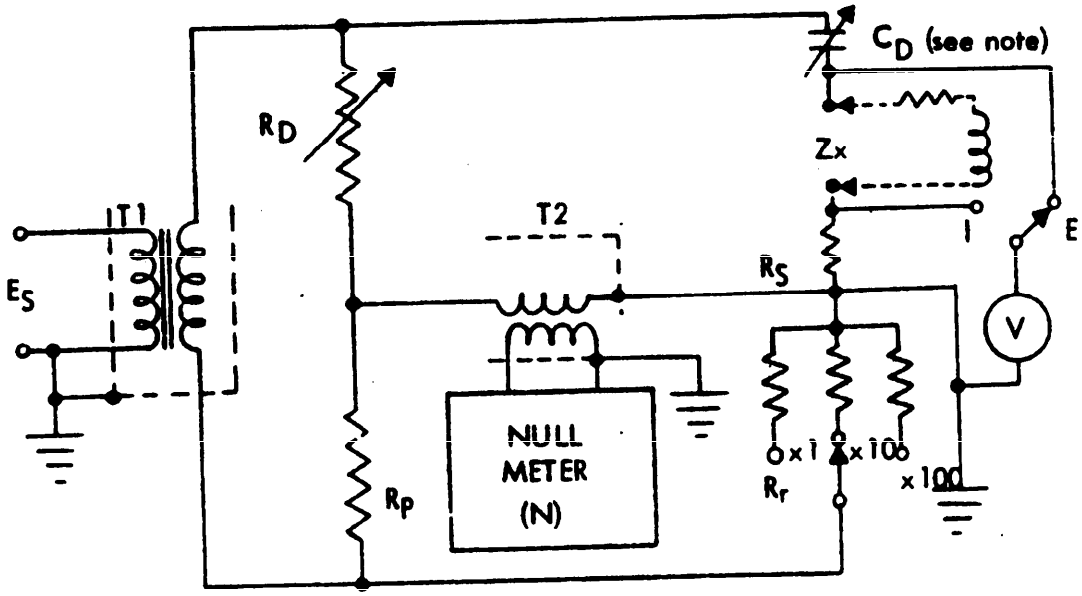
(f) Primary short-circuit impedance: 100 ohms maximum from 400 cps to 10 Kc

(g) Capacity from primary to shield: 15 pf maximum

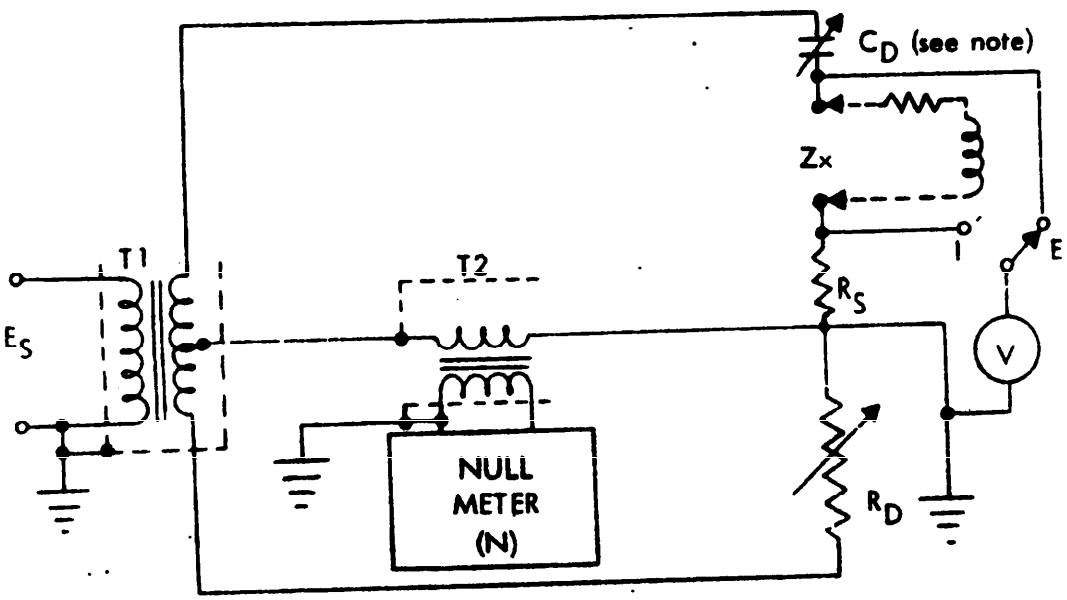
(h) The transformer must be resonated with 50 millivolts excitation at the operating frequency

20.3.8 Phase shifter (PS). - PS is a variable phase shifter which can be set to the nominal phase shift of the resolver under test.

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CIRCUIT A



CIRCUIT B

NOTE: If necessary, plug-in capacitors may be added in parallel with C_D (paragraph 20.1.3).

Figure A1. Resolver Impedance Bridge

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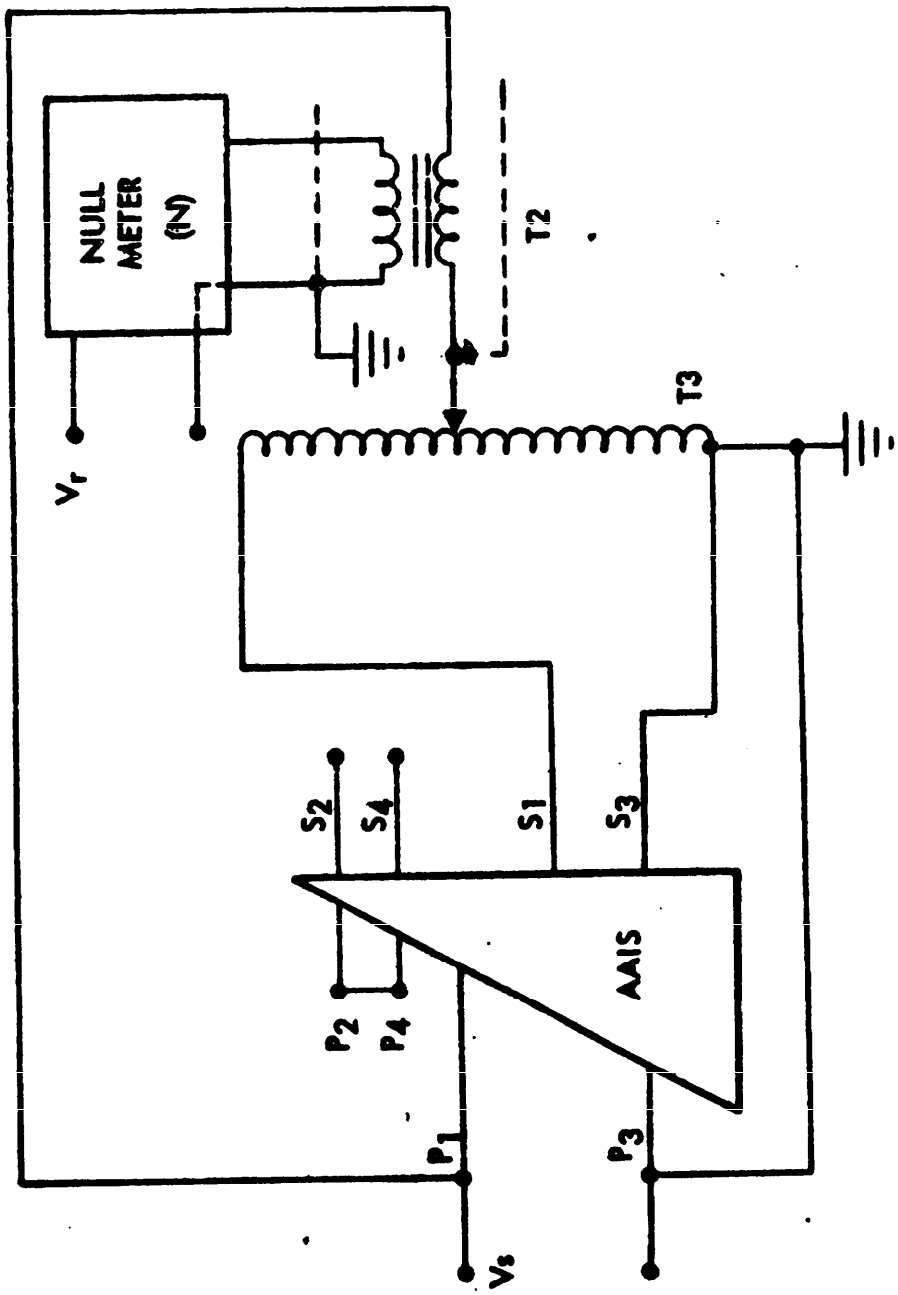


Figure A2. Transformation Ratio (K_t) Test - Greater Than Unity

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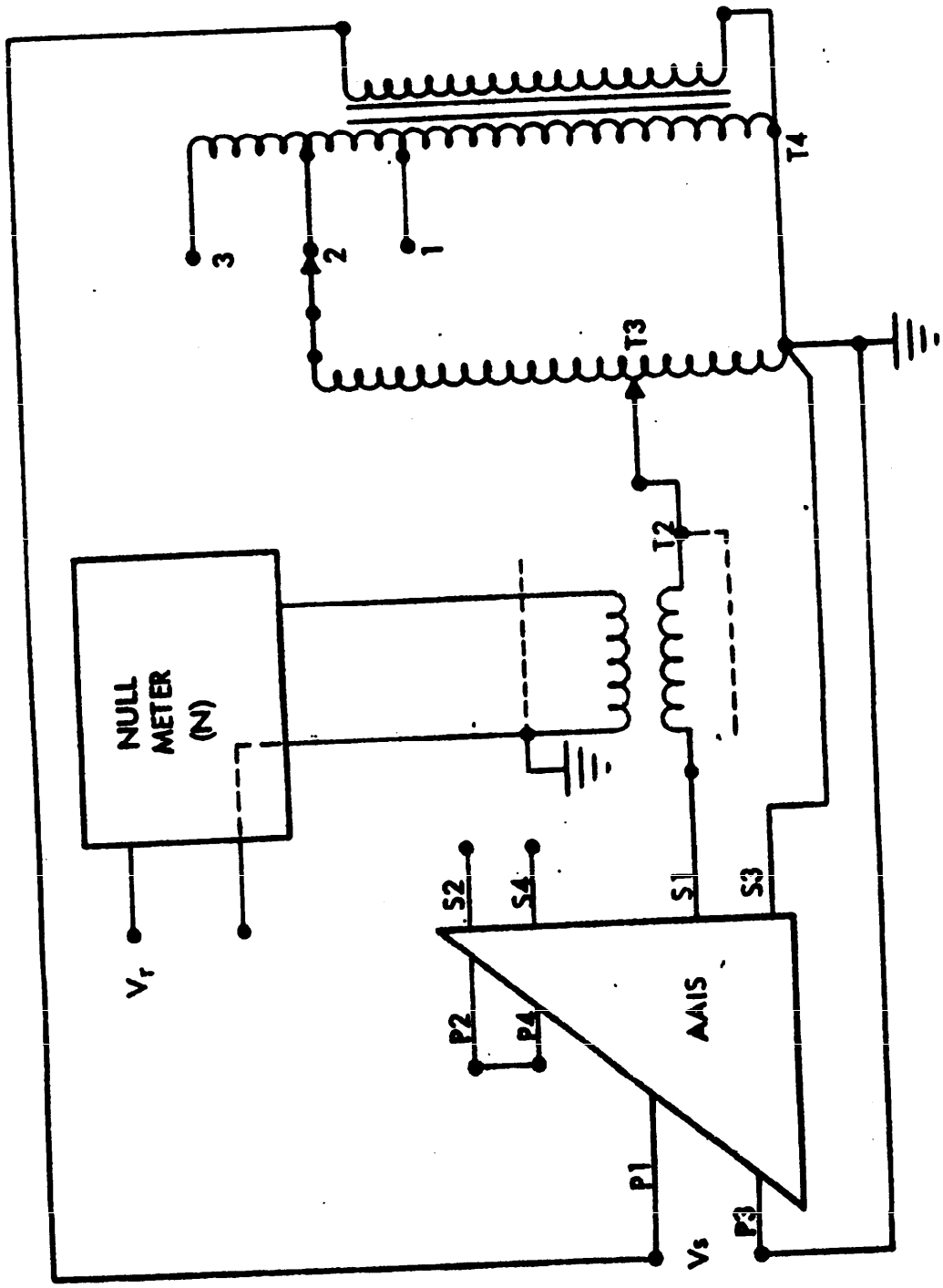


Figure A2A. Transformation Ratio (K_t) Test - Greater Than Unity (Alternate)

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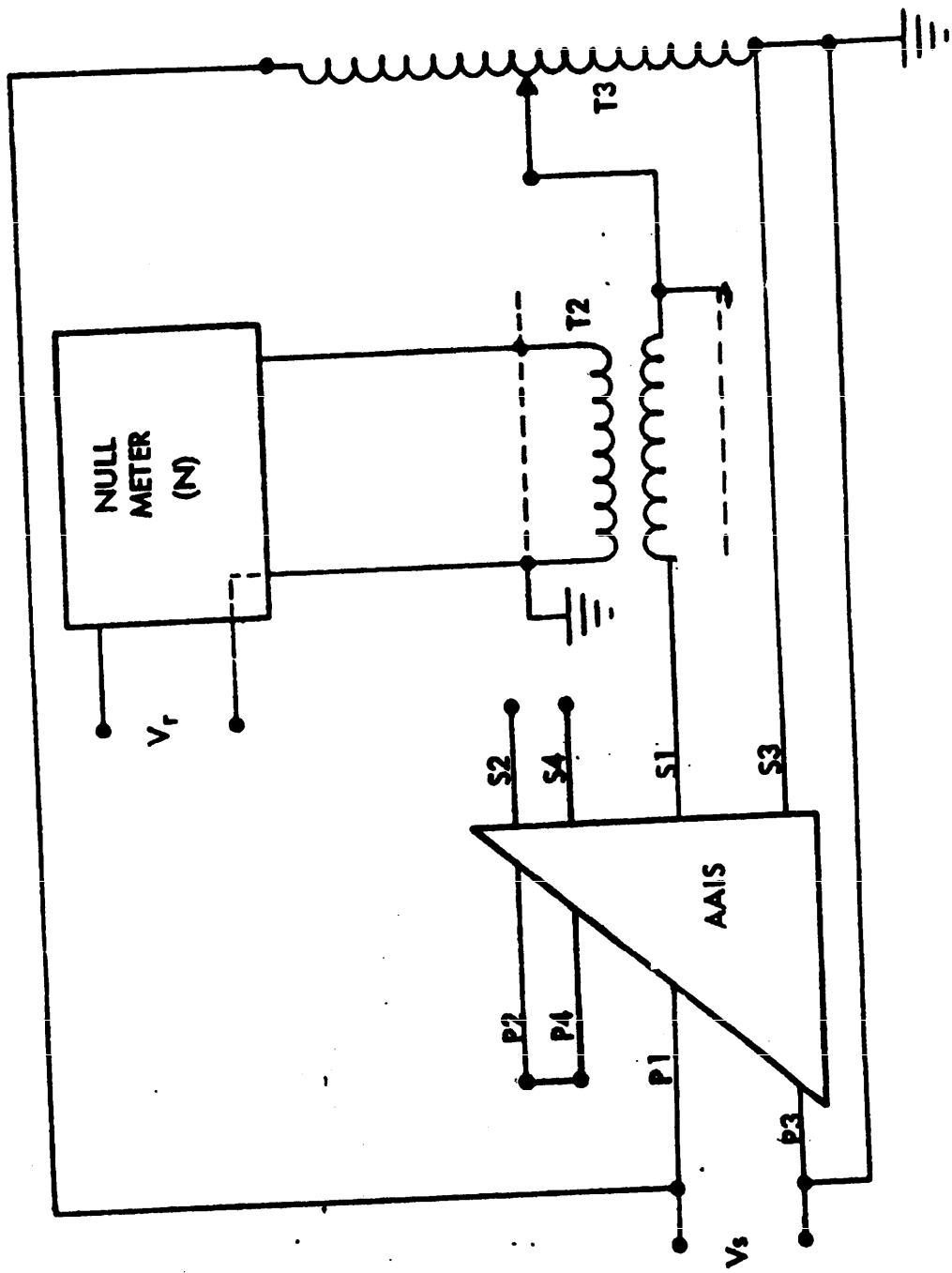


Figure A3. Transformation Ratio (K_1) Test - Less Than Unity

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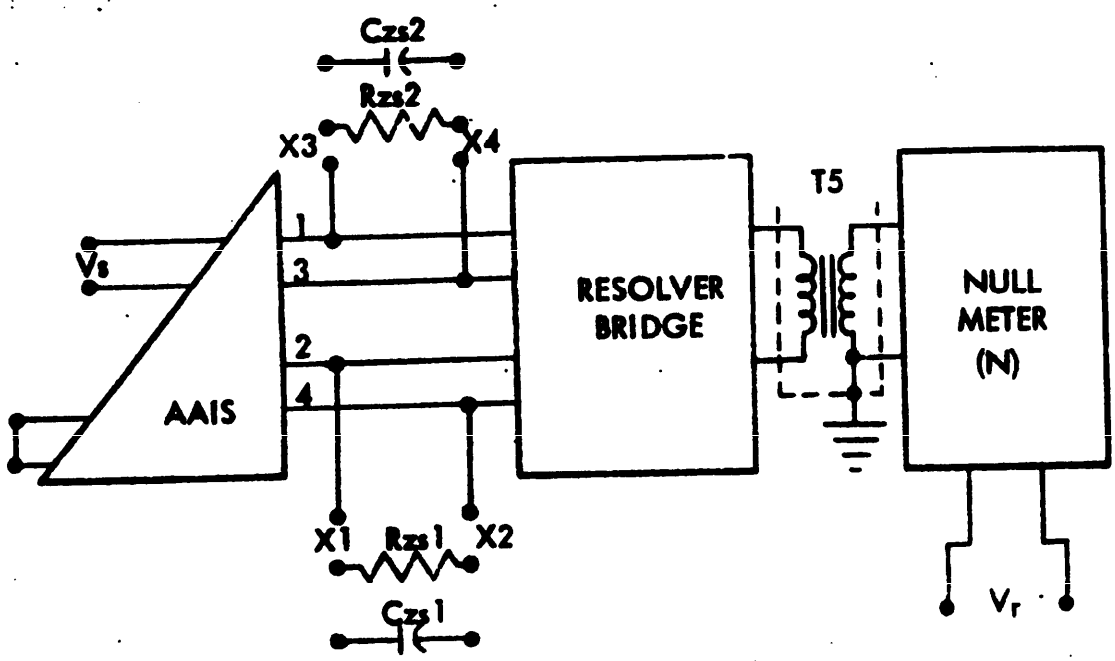
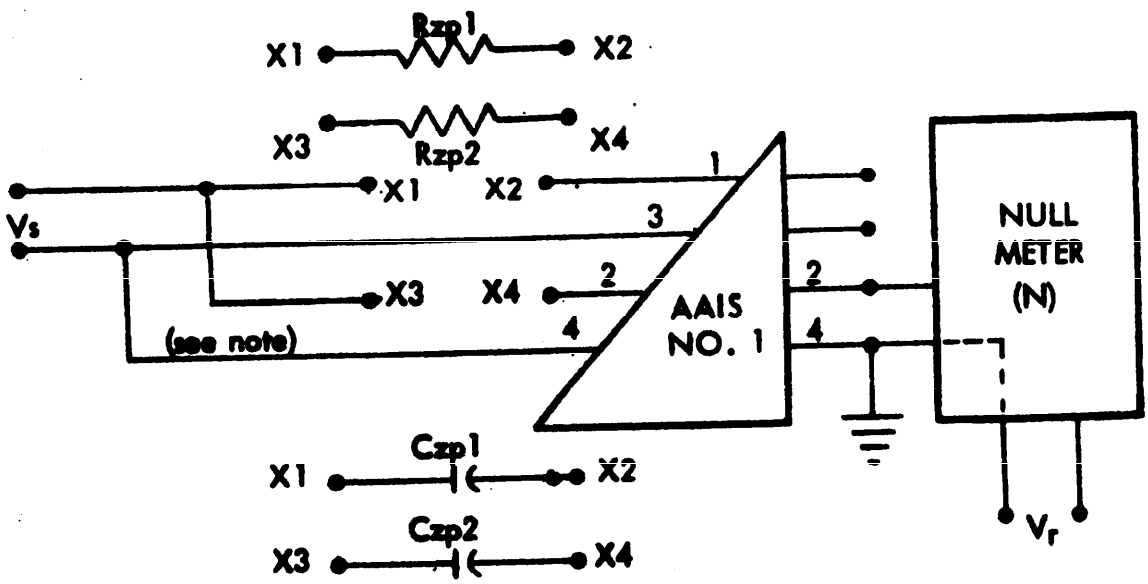


Figure A4. Impedance Balance Test - Secondary



Note: Parallel the input windings and reduce the input voltage to 0.707 Vs for 45 degree electrical excitation.

Figure A5. Impedance Balance Test - Primary

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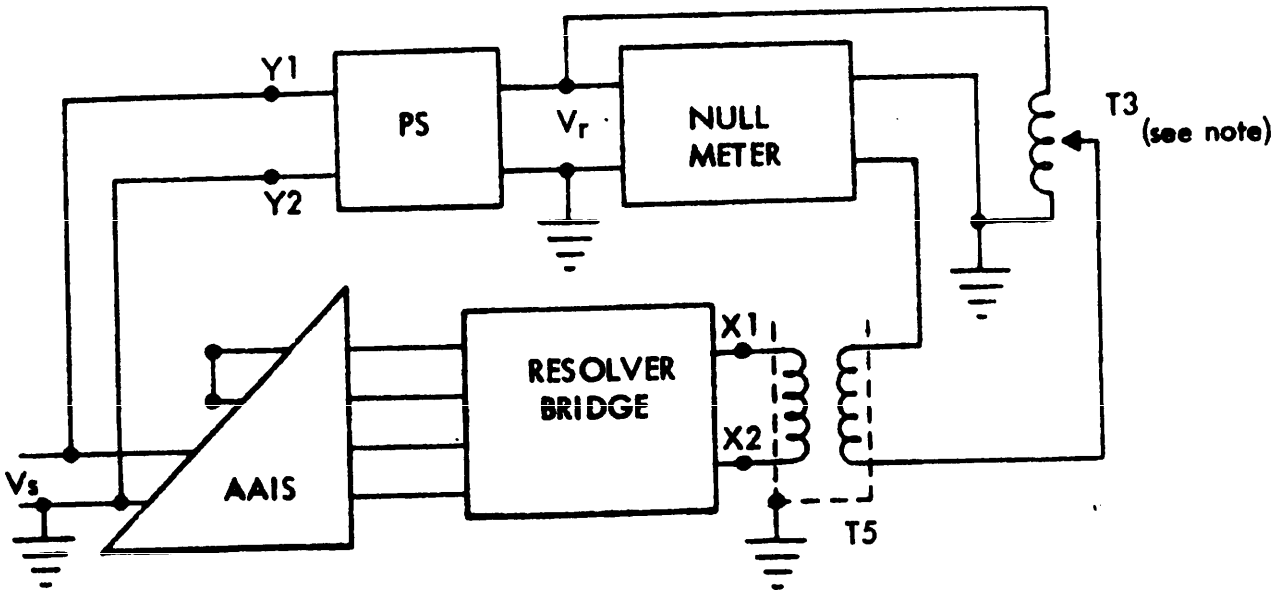
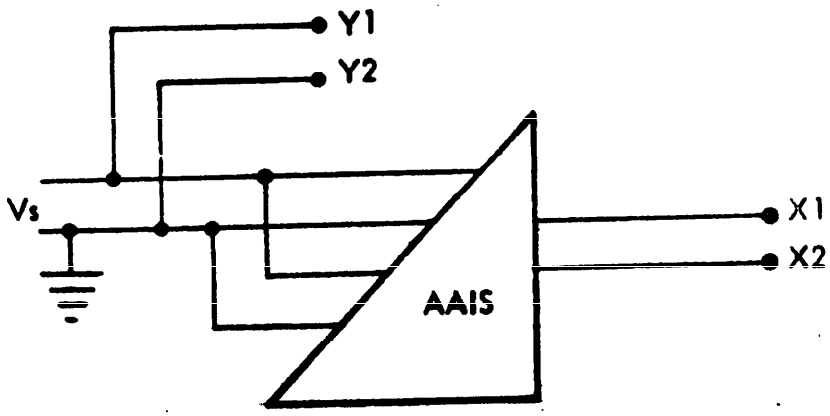


Figure A6. RX Resolver Connection, Environmental Tests



Note: Set T3 to zero for indication of any variation in electrical angle. Set T3 for zero in-phase voltage for total null measurement.

Figure A6A. RC Resolver Connection, Environmental Tests

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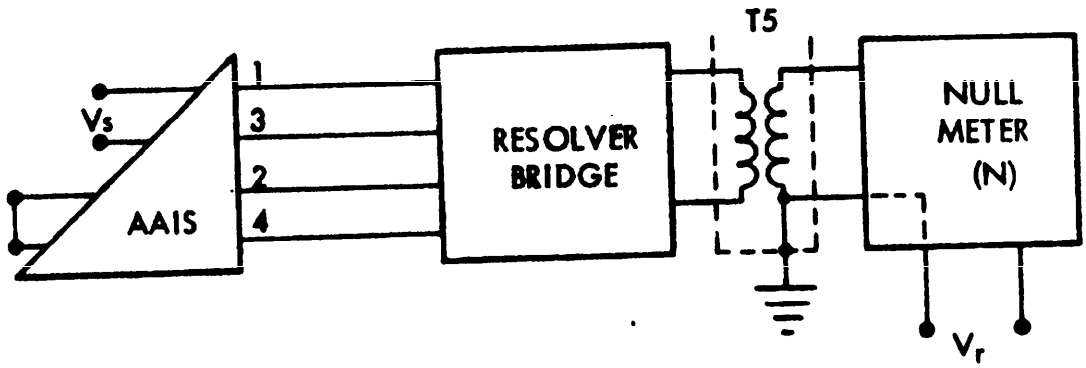
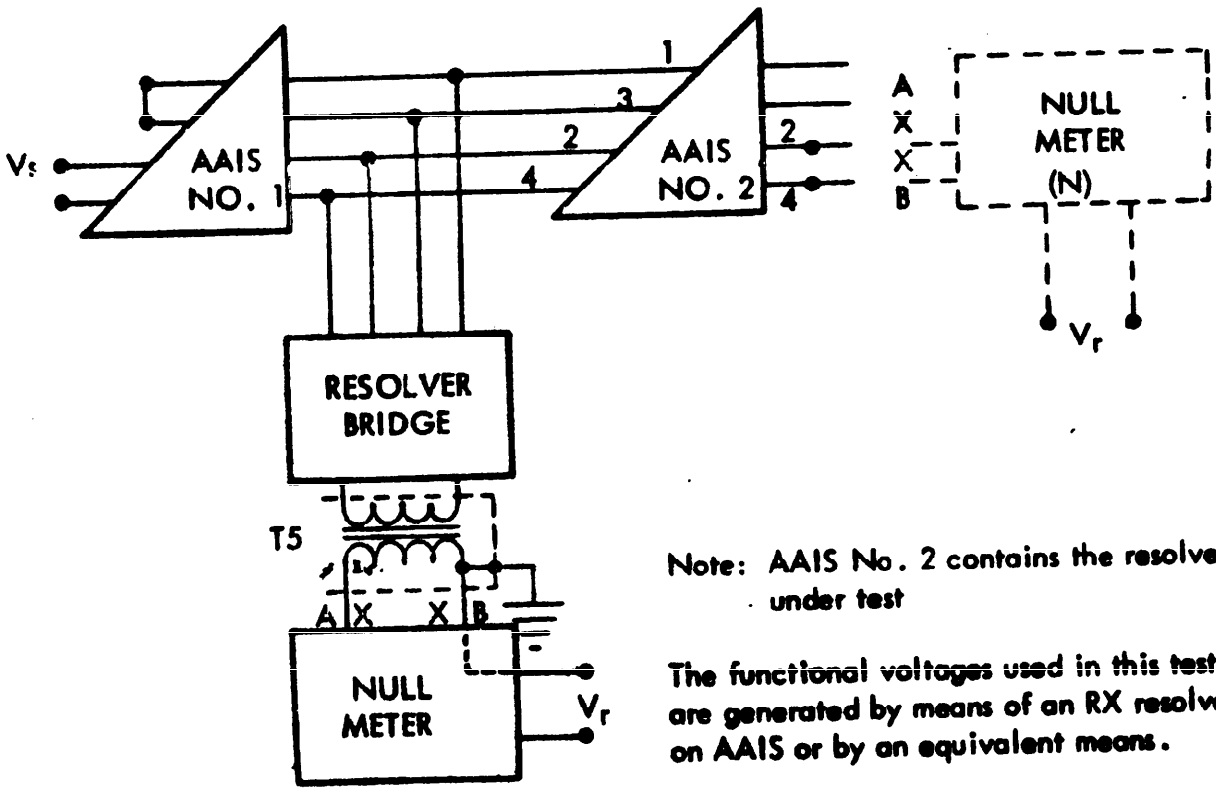


Figure A7. RX Resolver Angular Accuracy Test Connections



Note: AAIS No. 2 contains the resolver under test

The functional voltages used in this test are generated by means of an RX resolver on AAIS or by an equivalent means.

Figure A7A. RC Resolver Angular Accuracy Test Connections

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-21530D(WP) RESOLVERS, ELECTRICAL, AC FOR USE IN POLARIS FLEET UTILITY WEAPON SYSTEM GENERAL SPEC

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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